Almost sure continuity of the higher-dimensional LQG metric

Abstract

After fixing the median distance between two unit-separated points to be one, the higher-dimensional γ-Liouville quantum gravity metrics can be chosen continuously in γ on the same field, almost surely, throughout the subcritical interval and at its Euclidean endpoint. We use the supplied uniqueness and strong metric characterization as a standing input. Exact scaling lets us compare two continuum metrics directly. A local comparison improves a global comparison in distribution; repeating this implication at fixed confidence avoids an iteration of exceptional events.

1 The theorem

Let $d \geq 3$, let $e_1 = (1, 0, \dots, 0)$, and let h be the whole-space log-correlated Gaussian field on $\mathbb{R}^d$, with average zero on the unit sphere. The logarithmic covariance has coefficient one: for smooth zero-integral test functions f, g,

$$\mathrm{Cov}((h,f),(h,g)) = \iint_{\mathbb{R}^d \times \mathbb{R}^d} f(x) g(y) \log \frac{1}{|x-y|}\, dx\, dy\,. \tag{1.1}$$

We take the higher-dimensional metric characterization and uniqueness theorem in the supplied manuscript [8, Theorems 1.2 and 1.4 and Section 10] as an established project input. In particular, the γ-LQG metric is a measurable function of h, unique up to a deterministic factor, and satisfies locality, the length-space property, Weyl scaling and exact similarity covariance. We also use that manuscript's identification of the fixed-parameter LFPP limit and its uniqueness of the unit-distance median.

Theorem 1.1. *Normalize the γ-LQG metric by*

$$\operatorname{med} D_h^\gamma(0, e_1) = 1, \qquad 0 < \gamma < \sqrt{2d}, \tag{1.2}$$

and set $D_h^0(x,y) = |x-y|$. The metrics admit versions on the same field such that, almost surely,

$$[0, \sqrt{2d}) \ni \gamma \longmapsto D_h^\gamma \quad \textit{is continuous for locally uniform convergence of distances.} \tag{1.3}$$

Thus, on one event of probability one, for every $\gamma_0 \in [0, \sqrt{2d})$ and every $R < \infty$,

$$\lim_{\gamma \to \gamma_0} \sup_{x,y \in \overline{B}_R} |D_h^\gamma(x,y) - D_h^{\gamma_0}(x,y)| = 0\,. \tag{1.4}$$

At $\gamma_0 = 0$ this says that the distances converge to Euclidean distances. At a positive γ_0 it says that all distances in a fixed bounded region change uniformly little when γ changes little.

The median convention in (1.2) is the higher-dimensional counterpart of Kavvadias's normalization in [7, Theorems 1.5 and 1.7]; it is also the point-distance convention of [2, (1.4)]. Kavvadias's conclusions are convergence in probability on the common field. The common almost-sure version in Theorem 1.1 requires additional control of parameter oscillations. Uniqueness identifies a limit at each fixed parameter; it does not provide that control.

Why the proof is shorter. The proof works with the canonical continuum metrics. Consequently there are no two cutoff scales to compare and no cutoff normalizers to calibrate. After dividing each metric by its exact scale factor, the joint law of a pair of metrics is the same at every radius. We first obtain a local improvement of a metric comparison by a finite family of smooth field changes. Annular sewing turns this local information into a new comparison at the original confidence level. The preceding comparison is used only to estimate a probability, so its exceptional event need not be retained at the next step. This is the simplification that removes the former backward recursion of domains and additive errors.

The remaining quantitative estimates have two purposes. They construct a continuous version on compact positive parameter intervals, and they prevent exceptional narrow spikes as the parameter tends to zero. For the latter purpose we use the supplied continuum small-parameter estimates [9, Section 3 and (7.4)], followed by a finite parameter interpolation. The almost-sure LFPP theorem supplied with the project [10] fixes the parameter and is not used as a uniform statement.

Notation in the proof. It is convenient to use the Weyl exponent $\xi = \xi(\gamma)$, characterized by

$$Q(\xi) = \frac{d}{\gamma} + \frac{\gamma}{2}, \qquad \phi(\xi) = \xi Q(\xi), \qquad 0 < \xi < \xi_c. \tag{1.5}$$

The map $\gamma \mapsto \xi$ is a continuous increasing bijection of the subcritical intervals, and $\gamma/\xi \to d$ at zero [3, Proposition 1.8 and Corollary 1.9]. Only within the proof, write

$$\rho_h^\xi = \exp\{-\mathbb{E}\log D_h^{\gamma(\xi)}(0, e_1)\}\, D_h^{\gamma(\xi)}, \qquad \mathbb{E}\log \rho_h^\xi(0, e_1) = 0. \tag{1.6}$$

This is a deterministic scalar change, not another metric model. It makes a comparison of two log unit distances centered, which is useful once in the proof. We return to the median normalization at the end.

For $R \geq 1$ let $\|F\|_R = \sup_{x,y \in \overline{B}_R} |F(x,y)|$. An internal metric is denoted by $\rho(\cdot,\cdot;U)$. Constants may depend on d and a fixed compact subcritical interval J. When J contains zero, these constants are chosen on J before a lower bound $a > 0$ is introduced. A polynomial in a^{-1}, v^{-1} always has a fixed finite degree and fixed coefficients. Its value may be increased when a finite list of requirements is combined.

2 The estimates used in the comparison

We need estimates for one metric, uniformly over a compact parameter interval. They do not assert continuity of the metric in that parameter. The uniformity is recorded separately because it is not a consequence of uniqueness.

Lemma 2.1 (Uniform metric estimates)**.** *Fix $J \Subset [0, \xi_c)$. There are $p, p_{\mathrm{esc}} > 0$, $0 < \sigma < 1 < \beta$ and $C, A < \infty$ with the following properties. For each deterministic $\xi \in J$ and $R \geq 1$, the metric*

ρ_h^ξ *has a random constant* $H_R \geq 1$ *such that*

$$\mathbb{E}H_R^p \leq C(1+R)^A, \tag{2.1}$$

$$\operatorname{diam}_{\rho_h^\xi}(B_s(z); B_{2s}(z)) \leq H_R s^\sigma, \quad z \in B_R, \quad 0 < s \leq 1, \tag{2.2}$$

$$\rho_h^\xi(x,y) \geq H_R^{-1}|x-y|^\beta, \quad x,y \in B_R. \tag{2.3}$$

Fixed compact annular crossings have inverse moments of every prescribed finite order; fixed buffered internal diameters have a positive moment; and fixed nonzero point distances have all finite logarithmic moments. For a prescribed fixed factor K*, compact competitors for all pairs in* B_R *cost less than* K^{-1} *times a mandatory crossing to* B_{SR}^c*, outside probability* $C_K S^{-p_{\mathrm{esc}}}$*. The fixed-parameter metrics are proper and geodesic.*

The function ϕ *is locally Lipschitz on* $[0, \xi_c)$*,* $\phi(0) = 1$*, and*

$$\inf_{\xi \in J}\{\phi(\xi) - \sqrt{2d}\,\xi\} > 0. \tag{2.4}$$

Near zero, every prescribed finite list of moment orders, upper Hölder exponents less than one, lower exponents greater than one, and escape powers is available after shrinking the parameter interval. All expectations here precede the supremum over deterministic parameters.

Proof. Appendix A proves the uniform finite-approximation estimates from the DGZ quantile argument. For each fixed ξ, the supplied characterization [8, Corollary 10.4] identifies its limit in probability with the canonical metric. The positive and inverse moments make the log unit distances uniformly integrable, so mean-log normalization converges to (1.6). Take an almost-sure subsequence for this one fixed parameter and apply Fatou to compact crossings and buffered competitors. The bounds in Appendix A are independent of that parameter, which gives the displayed marginal estimates. Interior competitors and first-exit barriers give the same passage for internal distances. No common event over parameters is asserted in this argument.

For local Lipschitz continuity, a finite-range field h_n satisfies $\mathbb{E}\sup_V |h_n| \leq C_V(n+1)$ on a fixed box. If F_n^ξ is its unnormalized internal unit distance, the pathwise comparison bounds $|\log F_n^\xi - \log F_n^\eta|$ by $|\xi - \eta| \sup_V |h_n|$. Thus $(n \log 2)^{-1}\mathbb{E}\log F_n^\xi$ is Lipschitz in ξ, with a common constant for $n \geq 1$. Its pointwise limit is $\phi(\xi) - 1$. Taking the limit preserves that bound, including $\xi = 0$. The strict gap follows from the definition of the continuous subcritical regime and compactness. The small-parameter strengthening follows either from the last corollary in the appendix or directly from [9, Proposition 3.2]. □

For most of the comparison it is useful to pin the field on the sphere of radius ten. If h is initially pinned on the unit sphere, put $g = h - h_{10}(0)$. Then

$$\rho_h^\xi = e^{\xi h_{10}(0)}\rho_g^\xi, \qquad \mathbb{E}\log \rho_g^\xi(0, e_1) = 0. \tag{2.5}$$

The second identity uses the centered Gaussian pin. All estimates in Lemma 2.1 hold for this pin too, by Hölder's inequality and the exponential moments of the difference of the two fixed pins. We work under this radius-ten pin until the final construction.

For any radius r and center z, exact scaling gives

$$\rho_h^\xi(z + rx, z + ry) = S_\xi(r,z)\rho_{h^{r,z}}^\xi(x,y), \qquad S_\xi(r,z) = r^{\phi(\xi)}e^{\xi h_r(z)}, \quad h^{r,z} = h(z + r\,\cdot) - h_r(z). \tag{2.6}$$

The pair $(\rho_{h^{r,z}}^\xi, \rho_{h^{r,z}}^\eta)$ has the same joint law as the pair on the unit-sphere-pinned field, for every r, z. The same identities hold with $h_r(z)$ replaced everywhere by $h_{10r}(z)$; the transformed field

then has the radius-ten pin. We write $S_\xi^{10}(r,z)$ for that choice. For $\tau=|\eta-\xi|$ the physical scale ratio is nevertheless

$$\frac{S_\eta(r,z)}{S_\xi(r,z)}=\exp\{(\phi(\eta)-\phi(\xi))\log r+(\eta-\xi)h_r(z)\}. \tag{2.7}$$

This factor must be retained when a local comparison is used in physical coordinates.

Lemma 2.2 (Finite scale and pin bounds)**.** *Consider geometrically spaced radii in $[e^{-n},1]$, their own-radius center grids in B_W, and fixed enlargements of those grids. If $\log(1+W)\le c_0 n$, there are $C,c>0$ such that*

$$\max_{r,z}|h_r(z)|\le Cn \tag{2.8}$$

outside probability Ce^{-cn}. On this event, $|\log(S_\eta/S_\xi)|\le C\tau n$. Changing between the radius-one and radius-ten pins at a fixed unit inspection changes a comparison coefficient by $e^{(\eta-\xi)Z}$, where Z is a centered Gaussian of fixed variance. It multiplies an additive error, in primary units, by at most $e^{b|Z|}$, where $b=\sup J$.

Proof. There are at most Ce^{Cn} inspected center-radius pairs. The variance of each spherical average is at most $C(1+\log r^{-1}+\log(1+|z|))\le Cn$. A Gaussian tail at a sufficiently large multiple of n, followed by a union bound, proves (2.8). Equation (2.7) and the Lipschitz bound for ϕ prove the scale-ratio assertion. The last assertion follows by applying the constant Weyl factor separately to the two parameters. We use a fixed large bound for $|Z|$ when only fixed confidence is required. These pin caps are not inserted into local annular events. □

2.1 Local observations and geodesics

For an open set U, let $\mathcal{G}(U)$ be the closed Gaussian space generated by zero-mass smooth tests supported in U. Its covariance norm is a fixed multiple of $\dot H^{-d/2}$, and the Cameron–Martin norm of a shift, modulo constants, is the corresponding multiple of $\dot H^{d/2}$. Locality implies that the internal metric on an annulus, divided by its own factor $S_\xi(r,z)$, is measurable from the restriction of $h-h_r(z)$ to that annulus. The sphere used for this pin belongs to a fixed enlarged annulus. Mollification of the sphere measure puts these observations in its closed Gaussian space.

Lemma 2.3 (Gaussian block inequality)**.** *For an inner ball of radius r and a fixed-shape annulus of radius $R\gg r$, their zero-mass Gaussian spaces have operator correlation at most Cr/R. Equal-size windows at separation Ar have correlation at most CA^{-2}. At a sufficiently large fixed geometric spacing, events F_j in the annular blocks satisfy*

$$\mathbb{P}\Big(\bigcap_j F_j\Big)\le\prod_j\mathbb{P}(F_j)^{1/2}. \tag{2.9}$$

The same holds for suitably separated trial windows on a line.

Proof. Scale the outer window to unit size and put $q=r/R$. After subtracting the value of the kernel at the inner center, every required derivative of the cross kernel on the two fixed separated supports is $O(q)$. Duality with $\dot H^{-d/2}$ in each variable, using a fixed integer Sobolev order above $d/2$, bounds the operator correlation by Cq. For equal-size windows subtract in both variables; the corresponding derivative bound is $O(A^{-2})$. Thus the sum of the off-diagonal block operator norms in each row is at most one at sufficiently large spacing. The full covariance is bounded by twice its block diagonal. The Gaussian product inequality [5, Theorem 1(i)] with all exponents two gives (2.9) for finite-dimensional projections. Conditional expectations and bounded L^2 convergence give it for the closed Gaussian spaces. □

Here is the covering consequence we will use. Suppose a local event, in a fixed enlarged annulus, has failure at most p at every inspected radius. Own-radius grids in B_W and $m \asymp V$ separated radii in $[e^{-2V}, e^{-V}]$ then give

$$\mathbb{P}\{\text{some point of } B_W \text{ has no good central-tenth ball}\} \le C(1+W)^d e^{2dV} p^{m/2}. \tag{2.10}$$

Indeed, take anchors of mesh ce^{-2V} and use their nearest grid center at each radius. The tests remain annular about the anchor. Apply (2.9), then sum over the anchors. Choose p once sufficiently small to make the right side $C(1+W)^d e^{-cV}$. All pin caps used to convert units are paid separately.

Lemma 2.4 (A selector unchanged by scalar multiplication)**.** *On the space of proper length metrics on $\mathbb{R}^d$ inducing its usual topology, there is a jointly Borel choice $P(D,x,y) \in C([0,1],\mathbb{R}^d)$ of a geodesic at constant relative metric speed. For every $c > 0$,*

$$P(cD,x,y) = P(D,x,y).$$

Proof. For $L = D(x,y)$ the midpoint set

$$M = \{z : D(x,z) = D(z,y) = L/2\}$$

is nonempty and compact. Near-minimizing paths supply approximate midpoints; properness and the triangle inequality give exact midpoints. Choose its lexicographically first point. Here is an explicit Borel construction. Put $F(z) = |D(x,z) - L/2| + |D(z,y) - L/2|$ and recursively set

$$a_j = \lim_{n\to\infty} \inf\{q_j : q \in \mathbb{Q}^d,\ F(q) < 1/n,\ |q_i - a_i| < 1/n \text{ for } i < j\}.$$

The feasible sets decrease and are nonempty by rational approximation. All their points lie in the proper ball $D(x,q) < L/2 + 1$. Compactness therefore proves that each limit is finite and is the minimum of coordinate j on the preceding minimizing face of M. The construction is Borel and depends only on the exact midpoint set, which scalar multiplication does not change.

Apply this midpoint rule recursively at dyadic times. Triangle inequality in both directions gives $D(P(s),P(t)) = |s-t|L$ on dyadic times. All selected points lie in a compact metric ball, on which the identity to Euclidean space is uniformly continuous. They consequently have a unique continuous extension. Euclidean piecewise-linear interpolants of the dyadic values converge uniformly to it. Each interpolant is Borel, so the limit is Borel and retains scalar invariance.

For completeness, the metric domain is Borel in the compact-uniform profile topology. Triangle inequalities are tested at rational points; positivity off the diagonal is positivity of minima on the compact sets $\{|x|,|y| \le R,\ |x-y| \ge 1/k\}$. Properness is the countable condition $\forall m\ \exists R\ \forall q \in \mathbb{Q}^d \cap B_R^c : D(0,q) > m$. Approximate rational midpoints are a further countable condition, and, with properness, give exact midpoints for every pair and the preceding geodesic construction. Outside this Borel domain use the straight path as a harmless Borel fallback. The canonical metrics are Borel profiles by the supplied strong characterization, and bounded field shifts preserve properness. Thus the selector applies to every metric used in the finite Gaussian count. □

Put

$$A = \{1/2 < |x| < 7/2\}, \qquad K = \{3/4 \le |x| \le 1\}. \tag{2.11}$$

Fix high-probability local references: intrinsic lower Hölder bounds in A, upper bounds in doubled balls contained in A, and upper and lower bounds for the prescribed crossings and diameters.

These are local predicates; global estimates prove their probabilities. They imply that for one fixed $\delta_0 > 0$,

$$R(u, v; A) = R(u, v), \quad u, v \in K, \quad |u - v| \le \delta_0, \tag{2.12}$$

for either metric R, under every exterior extension of the field. To see this, choose δ_1 inside the buffer of K and then choose δ_0 so that the local competitor for such endpoints costs less than the lower bound for a first exit from a δ_1 ball.

For a pair in its separate unit normalizations, define $H_{\downarrow}(b)$ to be the existence of endpoints on $\partial B_{3/4}$ and ∂B_1 joined by an intrinsic E-geodesic in K, which is also minimizing in A, and for which $E_A \le bD_A$. Define $H_{\uparrow}(b)$ using an intrinsic D-geodesic in K minimizing in A, and $E_A > bD_A$. For $H_{\downarrow}$ the separate unit normalizations use the sphere-one pin; for $H_{\uparrow}$ they use the sphere-ten pin. Thus the latter block includes that outer sphere, but its shape is still fixed. This convention makes $H_{\uparrow}$ agree exactly with the law used in the unit-scale forcing count. Both are local Borel events. Here is a direct verification of the existential quantifier. View the observations as continuous intrinsic metric profiles on K^2. If these profiles converge uniformly to nondegenerate profiles, witness lengths are bounded by the maximum of the geodesic's own profile on K^2. For every Euclidean separation $t > 0$, the limiting profile has a positive minimum on $\{|x - y| \ge t\}$. Uniform convergence and constant relative arclength therefore give a common Euclidean modulus for the witnesses. Arzelà–Ascoli preserves their endpoint spheres, images in K, lengths and geodesic equalities at rational times. Existence with a weak ratio inequality is consequently closed relative to the domain of nondegenerate metric profiles. For $H_{\uparrow}$, take the countable union with $E_A - bD_A \ge 1/m$. That profile domain is Borel by rational triangle tests and positive minima on separated compact sets. Pulling these events back by locality proves the assertion.

Divide a witness into a fixed large number N_* of equal own-arclength pieces. The local references make every piece have Euclidean diameter less than $\delta_0/2$ and give fixed two-sided Hölder moduli in its normalized arclength. Triangle inequality shows that one piece retains the favorable, or unfavorable, ratio. Its own length has a fixed positive lower bound, so its endpoints have Euclidean separation at least a fixed $w_* > 0$, and its primary distance is at least $d_* > 0$. Equation (2.12) makes its endpoint distances global. In particular, a bad annular witness gives a separated global bad pair. A global additive comparison of size u on a fixed guarded box gives, by the same subdivision and the crossing lower bound, an annular multiplicative error at most Cu. These deterministic implications do not multiply the exceptional probability.

3 Comparison by local improvement

Fix two deterministic parameters $\xi, \eta \in J$, put $D = \rho^{\xi}$, $E = \rho^{\eta}$, and write $\tau = |\eta - \xi|$. This section first explains the annular argument. The finite field-change construction used for its local improvement is proved in Section 4.

3.1 Sewing local shortcuts

Lemma 3.1 (Path replacement)**.** *Suppose every point of a continuous finite-R-length path P is in the central tenth of a supplied ball with radius in $[r_-, r_+]$. The intrinsic sphere diameters below are taken in fixed buffered annuli. If each inner R-crossing is at least mS_R, each outer sphere has Q-diameter at most MS_Q, and $S_Q/S_R \le L$, there is a Q-path whose endpoints are within $2r_+$ of those of P and whose length is at most $2(M/m)L \operatorname{len}_R(P)$.*

Suppose instead that $P : [0, \ell] \to B_W$ is an R-geodesic, $Q \le CR + e$ on all its subsegments, and its global moduli are at most H. Suppose each supplied outer sphere has R-diameter at most MS_R,

its thin crossing is at least mS_R, and every actual thin R-block has a Q-connector costing at most b times its length. For $C \geq b$, put $\vartheta = m/(2M)$, $q_* = 1-\vartheta$, $B_- = H^{-1}(r_-/4)^\beta$ and $e_+ = 2H(2r_+)^\sigma$. Then

$$Q(P(0), P(\ell)) \leq \left[b + q_*(C-b) + \frac{e}{B_-} \right] \ell + Ce_+ + e. \tag{3.1}$$

The conclusion holds on every subsegment of the same original geodesic.

Proof. For each core visit retain the connected component of the path's preimage of its supplied ball. These intervals cover the parameter interval. A finite inclusion-minimal subcover has multiplicity at most two: in a triple overlap the middle interval is covered by the other two. The interval endpoints have the same order. Consecutive balls overlap and neither contains the other, so their boundary spheres intersect. Connect these intersection points in the supplied annuli, omitting the first and last connectors. Every retained intermediate interval contains a core visit and pays a crossing; the total crossing charge is at most twice the original length. This proves the first assertion.

For the second, at most two intervals are clipped by the endpoints, and their total length is at most e_+ by geodesicity. Every remaining interval has length at most MS_R and contains a thin block of length at least mS_R. Two-color the intervals and retain the color carrying more block length. Its disjoint blocks have total length $B \geq \vartheta(\ell - e_+)$ and number at most ℓ/B_-. Use the b-connectors on those blocks and the old comparison on the gaps. The resulting cost is $C\ell - (C-b)B + (\ell/B_- + 1)e$, which gives (3.1). All new comparison endpoints stay on the original geodesic. □

Lemma 3.2 (Rough comparison)**.** *There is a fixed $C_b > 1$ such that, for $R \geq 1$ and $e, q \in (0, 1/8)$,*

$$n \geq C\{1 + \log(1+R) + \log e^{-1} + \log q^{-1}\}, \qquad \tau n \leq c \tag{3.2}$$

imply

$$\mathbb{P}\{E \leq C_b D + e,\ D \leq C_b E + e \text{ on } B_R^2\} \geq 1 - q. \tag{3.3}$$

This holds under either of the two fixed sphere pins.

Proof. At radii between $e^{-u'n}$ and e^{-un}, with fixed $0 < u < u' < 1$, use the local crossing lower and sphere-diameter upper references for both own-normalized metrics. Their failure is a sufficiently small fixed number. Equation (2.10) supplies a ball at every point of B_W, outside $C(1+W)^d e^{-cn}$. The separate Gaussian cap in Lemma 2.2 bounds both physical unit ratios by two.

The one-metric escape estimates give all source geodesics from B_R inside a ball of radius W polynomial in $1 + R, q^{-1}$, outside $q/4$. Both global upper moduli there are bounded by another such polynomial, outside $q/4$. Apply the first assertion of Lemma 3.1 in each direction and attach the free endpoints by the upper modulus. Their cost is at most $CHe^{-\sigma un}$. The first condition in (3.2) pays that cost and all cover and cap failures. The comparison coefficient depends only on the fixed local references. □

We next isolate the amplification statement. It applies to either orientation of the pair and to either fixed pin in the local event. In the forward orientation the bad event can be $H_\uparrow(b)$; for the reverse orientation, the complement of $H_\downarrow(b^{-1})$ supplies the desired local connectors.

Proposition 3.3 (Annular amplification)**.** *There is a fixed $p_0 > 0$ with the following property. Suppose a local own-normalized event of failure at most p_0 guarantees that every actual primary*

thin-annulus block has a secondary connector costing at most b times its primary length, where $C_b^{-1} \le b \le C_b$. *Then there are fixed* $M, C, c > 0$ *such that, for* $R \ge 1$, $0 < \delta, e < 1/8$, *and*

$$n \ge C\delta^{-M}\{1 + \log(1+R) + \log e^{-1} + \log \delta^{-1}\}, \qquad \tau n \le c\delta, \tag{3.4}$$

one has

$$\mathbb{P}\{E \not\le (b+\delta)D + e \text{ on } B_R^2\} \le C_R \delta^{-M} e^{-c\delta^M n}. \tag{3.5}$$

Here D, E *denote the chosen primary and secondary metrics.*

Proof. Reserve slack in the rough constant. Lemma 3.2 gives a rough coefficient at most C_b, additive error $e^{-c_0 n}$ and failure $e^{-c_0 n}$ on any box whose logarithmic radius is a sufficiently small multiple of n. Fix $c_0 > 0$ first. Choose a fixed L so large that $4\beta/L < \min\{\sigma/4, c_0/16\}$, and put

$$N = \lceil C_\vartheta \log((C_b+1)/\delta)\rceil, \quad a_j = \tfrac{1}{4}L^{-j}\ (1 \le j \le N), \quad a_* = a_N. \tag{3.6}$$

The constant C_ϑ makes $q_*^N(C_b+1) \le \delta/16$. Use geometrically spaced radii in $[e^{-4a_j n}, e^{-a_j n}]$ in band j. The number of radii in each band is comparable to $a_j n$. In particular, $a_* \ge c\delta^{M_0}$ for a fixed M_0.

Choose small fixed $h > 0$ and then still smaller $w > 0$, and set

$$W = C_R e^{w a_* n}, \qquad H = C_R e^{h a_* n}. \tag{3.7}$$

The one-metric bounds give confinement to B_W and global moduli at most H, outside $C_R e^{-c a_* n}$, provided $Aw < ph/2$. Take w small enough also for the rough comparison on B_{W+2}. The annular covers hold in every band outside $C_R N e^{-c a_* n}$, after choosing p_0 sufficiently small and decreasing w. These covers use only the local own-normalized events. The separate finite Gaussian cap changes their physical connector coefficient from b to at most $be^{C\tau n} \le b + \delta/16$. Its failure is exponentially small in n.

Sew the bands from the smallest physical radii to the largest, always on subsegments of the same original primary geodesic. If a comparison coefficient is already smaller than the target local coefficient, retain it. Otherwise (3.1) contracts its excess by q_*. The first additive-error quotient is bounded by $CHe^{-(c_0 - 4\beta a_1)n}$. Subsequent quotients, including their fixed constants, are bounded by

$$CH^2 e^{-(\sigma a_j - 4\beta a_{j+1})n} \le CH^2 e^{-3\sigma a_j n/4}. \tag{3.8}$$

Take $h < \min\{\sigma/16, c_0/16\}$. The geometric sum of these quotients is at most $\delta/8$, and the last additive error is at most e, once the first inequality in (3.4) holds, with M increased. Contraction, the $\delta/16$ physical drift and the error quotient sum therefore give coefficient at most $b+\delta$. The total failure is $C_R N e^{-c a_* n}$, which has the form (3.5). There is no cutoff depth consumed by rescaling: every own-normalized local pair has exactly the original joint law. □

3.2 A fixed local improvement

We first show that favorable secondary arcs have a fixed positive probability. This uses centering, not a previously sharp comparison.

Lemma 3.4 (Favorable arcs)**.** *There are* $p_* > 0$ *and* $C < \infty$ *such that, for* $0 < v < v_0$ *and* $\tau \le cv^C$,

$$\mathbb{P}(H_\downarrow(1+2v)) \ge p_*. \tag{3.9}$$

After imposing the fixed local references with smaller failure, the fixed subdivision described in Section 2 supplies a favorable arc with own length in a fixed compact subinterval of $(0,\infty)$ *and fixed two-sided Hölder moduli.*

Proof. On the unit-sphere-pinned field, $Y = \log E(0,e_1) - \log D(0,e_1)$ has mean zero and bounded second moment. Cantelli's inequality gives

$$\mathbb{P}\{E(0,e_1) \le (1+v)D(0,e_1)\} \ge cv^2. \tag{3.10}$$

If (3.9) fails for sufficiently small p_*, the complement of $H_\downarrow(1+2v)$ and the local references supply, on every primary E-block, a D-connector of coefficient $(1+2v)^{-1}$. This is one radius-independent own-unit event. Apply Proposition 3.3 in that orientation with $\delta = v/100$, $e = v^{A+1}/100$ and failure smaller than $cv^2/4$. A depth polynomial in v^{-1} suffices; its drift budget follows from the stated condition on τ. Choose A first so that the inverse point moment gives $\mathbb{P}\{D(0,e_1) < v^A\} < cv^2/4$. On a nonempty intersection with (3.10), the amplified inequality would imply

$$D(0,e_1) \le \big((1+2v)^{-1} + v/100\big)(1+v)D(0,e_1) + v^{A+1}/100 < D(0,e_1), \tag{3.11}$$

for small v, a contradiction. The reference and subdivision conclusion follows after their fixed failures are chosen below $p_*/2$. □

Proposition 3.5 (Improvement of the unit comparison)**.** *Fix a sufficiently large guarded radius R_s and a sufficiently small $q_0 > 0$. For $0 < a \le \xi, \eta \in J$ and $0 < v < v_0$, there are $g, \varepsilon_*, t_* > 0$ and a fixed polynomial P such that*

$$g \ge e^{-P(a^{-1},v^{-1})}, \quad g \le v, \quad 0 < \varepsilon_* \le v, \quad \log \varepsilon_*^{-1} \le P(a^{-1},v^{-1}), \quad t_* \ge e^{-P(a^{-1},v^{-1})}. \tag{3.12}$$

For $\tau \le t_*$ and $1 + 20v \le c \le C_b$, the implication

$$\mathbb{P}\{E \le cD + \varepsilon_* \text{ on } B^2_{R_s}\} \ge 1 - q_0 \quad \Longrightarrow \quad \mathbb{P}\{H_\uparrow(c-g)\} \le p_0/2 \tag{3.13}$$

holds. The source probability is under the radius-ten pin, exactly as in $H_\uparrow$.

The proof is given in Section 4. Its main point is that all geometric references, the source box and the target local failure are fixed before v. The finite count gives a gain $g \ge \exp[-P(a^{-1},v^{-1})]$. The same exponentially small ε_* suffices for every c in the displayed interval.

Proposition 3.6 (A seed at fixed confidence)**.** *For a fixed guarded ball B_{R_s} and a fixed sufficiently small $p_s > 0$, there is $B < \infty$ such that*

$$\tau \le \exp[-Ca^{-B}u^{-B}] \quad \Longrightarrow \quad \mathbb{P}\{\|D - E\|_{R_s} > u\} \le p_s, \qquad 0 < u < 1. \tag{3.14}$$

The constants are uniform for $0 < a \le \xi, \eta \in J$.

Proof. Take a uniform fixed upper quantile M_s for both diameters on B_{R_s}, with combined failure at most $p_s/2$, and put $v = \min\{v_0/2, u/(160M_s)\}$. Choose $q_0 \le p_s/8$ in Proposition 3.5. Decrease ε_*, if necessary, to at most $u/16$; its logarithmic bound remains polynomial.

Lemma 3.2 starts with coefficient C_b, error ε_* and failure at most q_0. Suppose the current coefficient is $c \ge 1 + 20v$. Proposition 3.5 gives the rare local event $H_\uparrow(c-g)$. Add the already fixed local-reference failures within the spare $p_0/2$. Proposition 3.3, with $\delta = g/2$, returns

$$\mathbb{P}\{E \le (c - g/2)D + \varepsilon_* \text{ on } B^2_{R_s}\} \ge 1 - q_0. \tag{3.15}$$

A depth at most $Cg^{-M}(1 + \log \varepsilon_*^{-1} + \log q_0^{-1})$ suffices. It is at most $\exp[P(a^{-1},v^{-1})]$. One parameter ceiling $\tau \le \exp[-P'(a^{-1},v^{-1})]$ pays its drift budget and every condition of the local improvement.

Iterate this deterministic implication until the coefficient is at most $1 + 20v$. There are at most $2C_b/g + 1$ iterations. Each conclusion has the same error and confidence as its premise. No exceptional events are intersected across iterations, so neither the probability loss nor the error accumulates. Run the same argument with the metrics interchanged. Intersect only the two final comparisons and the diameter event. Their total failure is at most p_s, and their difference is less than u. A fixed power majorizes the finite polynomials in the parameter ceiling, giving (3.14). □

3.3 The parameter increment estimate

Proposition 3.7 (Increment bound). *For each $R < \infty$ there are $B, M, C, c > 0$ such that*

$$\mathbb{P}\{\| \log(1 + D) - \log(1 + E)\|_R > \delta\} \le C\delta^{-M} e^{-c\delta^M n} \tag{3.16}$$

whenever $0 < \delta < 1/8$, $0 < a \le \xi, \eta \in J$, and

$$\tau \le e^{-Ca^{-B}\delta^{-B}}, \qquad n \ge C\delta^{-M} \log(C/\delta), \qquad \tau n \le c\delta. \tag{3.17}$$

If, for this pair, the seed in (3.14) at accuracy $c\delta$ is already known by another argument, the first condition is unnecessary.

Proof. Use the seed at accuracy $c'\delta$ and the fixed local subdivision. The local references have fixed positive crossing lower bounds, so outside a sufficiently small fixed failure the annular comparison coefficient is $1 + c''\delta$ in each orientation. Apply Proposition 3.3 twice with slope allowance and additive error fixed small multiples of δ. It gives $E \le (1 + \delta/2)D + \delta/2$ and its reverse. Thus both ratios $(1 + E)/(1 + D)$ and its reciprocal are at most $1 + \delta/2$, proving (3.16). The seed is the only use of the first condition in (3.17). □

4 Local field changes

Throughout this section, the local metric reference constants are fixed before the tolerance $v \in (0, v_0]$. They may depend on a fixed compact subcritical parameter interval. A quantity called polynomial below is bounded by a fixed power of v^{-1}; Cameron–Martin energies have an additional factor ξ^{-2}. No Euclidean approximation of either metric is assumed. The primary and secondary metrics are the canonical strong metrics with parameters ξ and η; write $\tau = |\eta - \xi|$. Their Weyl changes are exact. We keep a small mismatch allowance Ω to compare the two different Weyl weights. The field-modification count follows the general comparison strategy of [4, Section 4]; the finite routed network below supplies its higher-dimensional geometric input, and a Gaussian overlap estimate replaces a product of local change-of-measure bounds. For a set K, write $K^{+t} = \{x : \operatorname{dist}(x, K) \le t\}$.

4.1 Quantitative routing and finite labels

Lemma 4.1 (Clearance around an arc). *Suppose an injective arc $\gamma : [0, 1] \to \overline{B}_1$ satisfies*

$$|\gamma(s) - \gamma(t)| \le A\,|s - t|^{\alpha}, \qquad |s - t| \le A\,|\gamma(s) - \gamma(t)|^{\beta}, \quad 0 < \alpha, \beta \le 1. \tag{4.1}$$

If $d \ge 2$, two points at distance at least $u \le 1$ from the arc can be joined at distance at least $c_A u^{1/(\alpha\beta)}$ from it. If the arc and the two points lie in a fixed convex box whose boundary has a fixed buffer from the arc, the joining path can stay in that box. It can be chosen polygonal with at most $C_A u^{-C_A}$ edges.

Proof. The inverse parameter on the arc extends to a β-Hölder function $F : \mathbb{R}^d \to [0,1]$ by the McShane formula for the metric $|x-y|^\beta$, followed by clipping to $[0,1]$. Thus $R(x) = \gamma(F(x))$ is a retraction. If $\operatorname{dist}(x,\gamma) \le r$, comparison with a nearest arc point gives

$$|x - R(x)| \le r + A^{1+\alpha} r^{\alpha\beta} \le C_A r^{\alpha\beta}.$$

The straight homotopy from x to $R(x)$, followed by contraction along γ, contracts the inclusion $\gamma^{+r} \hookrightarrow \gamma^{+C_A r^{\alpha\beta}}$.

For completeness, use finite cubical thickenings $\gamma^{+r} \subset P \subset \gamma^{+2r}$ and $\gamma^{+C_A(2r)^{\alpha\beta}} \subset Q \subset \gamma^{+2C_A(2r)^{\alpha\beta}}$, with $P \subset Q$. The inclusion $P \to Q$ is null homotopic. Natural Alexander duality for these compact polyhedra identifies the resulting zero restriction map on reduced cohomology in degree $d-1$ with the map

$$\widetilde{H}_0(S^d \setminus Q) \longrightarrow \widetilde{H}_0(S^d \setminus P)$$

induced by inclusion; see [6, Section 3.3]. Consequently all points outside Q belong to the component of infinity. Since $d \ge 2$ and P is compact, the complement of a ball containing P is connected, so these finite points belong to one component of $\mathbb{R}^d \setminus P$. This open component is path connected. Choose $r = c_A u^{1/(\alpha\beta)}$ so that $Q \subset \gamma^{+u/2}$.

Coordinate-clamping a path to the fixed box preserves the required clearance: every changed point lands on a boundary uniformly separated from the arc. Approximate at mesh $r/(100d)$ and erase repeated grid cubes. There are $O(r^{-d})$ cubes in the box; a polygonal path through them retains clearance $r/2$. Changing c_A proves the statement. □

The arc moduli arise from intrinsic geodesics without a lower bound on a secondary/primary unit ratio. Indeed, parameterize an intrinsic secondary geodesic of own length $\ell \in [\ell_0, \ell_1]$ by normalized own arclength. An intrinsic lower spatial bound gives the first modulus in (4.1). For the second, use the local internal upper Hölder bound for close pairs, whose doubled ball stays in the seed inspection; the fixed total length handles pairs farther apart. An ambient upper bound alone would not justify this step for an intrinsic geodesic. Use the own-length interval supplied by the fixed equal-arclength subdivision in Section 2; increase its fixed N_* once so that every piece has Euclidean diameter at most $\delta_0/2$, where δ_0 is the endpoint-separation threshold in (2.12), in seed coordinates. This diameter restriction is also imposed on every arc admitted by a label. Thus the local first-exit conclusion holds for every accepted witness, not only for the arc from which a label was first designed.

We record a precise finite version of the routing construction. Each trial has its fixed compact shell $K_{\mathrm{trial}} \Subset A_{\mathrm{seed}}$, with a deterministic positive buffer in seed units. Every source arc, including every arc admitted by a label, is required to have its entire image in K_{trial}. The favorable subarcs of the local subdivision in Section 2 already have this property. Take two rational ports on the boundary of a convex reserved box containing the trial and put K_{trial} in a smaller concentric box. All gate radii and their later enlargements are chosen smaller than this fixed seed buffer. Choose $t_0 = v^C$ so that its first and last $2t_0$ tails have the required small source diameter and secondary length. The initial endpoint has distance at least $(t_0/A)^{1/\beta}$ from $\gamma([t_0,1])$. Lemma 4.1 therefore gives a first lead avoiding this truncated arc by clearance $10r \ge cv^C$, and a second lead avoiding $\gamma([0,1-t_0])$ by the same clearance. Their numbers of edges are at most $N \le Cr^{-d}$. The subsequent perturbations move every point by less than r, leaving the weaker clearance r used in the source predicate.

Disjoint rational leads. Move free vertices slightly into the reserved box, then perturb them in smaller balls of radius $\delta \asymp r$. Round the endpoint attachments on a much finer rational grid. Use

a common mesh ν, chosen as a reciprocal integer multiple of the ports' fixed common denominator. Insert a free midpoint into every edge with two fixed endpoints. Every nonincident edge pair then has at most two fixed vertices among its four endpoints.

Choose free vertices successively. Exclude the lines through earlier or fixed vertex pairs, and exclude each affine plane which would make the last free vertex of an edge-pair quadruple coplanar with the other three. Those triples are noncollinear by the line exclusions. There are at most CN^3 forbidden subspaces at a step. A perturbation ball has $c(\delta/\nu)^d$ grid candidates, whereas a proper affine plane has at most $C(\delta/\nu)^{d-1}$; hence $\nu < c\delta/N^3$ suffices. For two nonincident edges, a nonzero three-coordinate determinant is an integer multiple of ν^3. Bounded edge vectors then give supporting-line distance at least $c\nu^3$. In $d > 3$, project to the three coordinates of a nonzero minor. The leads are simple and disjoint, with polynomial separation, when $d \geq 3$.

The retained straight middle segment of the original tree edge is deleted. Every free lead vertex has a controlled inward margin in its reserved box. A sufficiently thin lead tube can leave the box only near its own fixed port: its inward distance along the first segment grows at least linearly, with slope bounded below by the inward margin divided by the box diameter. Put this small neighborhood in the intended vertex region. It is separated from the opposite port and all other retained edges. Thus general position is not being used as a substitute for the port-incidence condition.

Covered-and-hit labels. Use closed grid cells of side μ in the reserved box. For a marked set of centers Q and a compact time interval I, require

$$\gamma(I) \subset \bigcup_{q\in Q} \mathrm{cell}(q), \qquad \gamma(I) \cap \mathrm{cell}(q) \neq \varnothing \quad (q \in Q). \tag{4.2}$$

A label records Q_* for $[0, 1]$, Q_0 for $[0, 2t_0]$, and Q_1 for $[1-2t_0, 1]$, as well as its trial and two rational lead sequences. Shared boundaries of closed cells do not matter. No avoidance of unmarked cells is required. The source existence predicate also requires the recorded first lead to have distance at least r from $\gamma([t_0, 1])$, the second to have distance at least r from $\gamma([0, 1-t_0])$, and their attachment endpoints to be within $d\nu$ of the corresponding arc endpoints. The fixed lead sequences obey the rational separation and reserved-port conditions above. Invalid finite configurations have empty source events. These requirements ensure that every allowed source witness, not just the witness first used to design a label, supplies the needed contacts and clearances.

Choose the rational mesh above sufficiently fine, before rounding the attachments, and then choose ω, μ so that

$$g = 100d\nu \ll r, \qquad A(10g)^\beta \ll t_0, \qquad \omega \ll \nu^3, \qquad \mu \ll \omega. \tag{4.3}$$

All are powers of v. The deterministic plateau chamber and gates are

$$C_e = \bigcup_{q\in Q_*} \overline{B}_\omega(q), \qquad G_0 = \bigcup_{q\in Q_0} \overline{B}_g(q), \qquad G_1 = \bigcup_{q\in Q_1} \overline{B}_g(q). \tag{4.4}$$

Writing $T_0 = \gamma([0, 2t_0])$, the exact label gives

$$T_0^{+(g-c_d\mu)} \subset G_0 \subset T_0^{+(g+c_d\mu)}, \qquad \gamma \subset C_e \subset \gamma^{+2\omega}, \tag{4.5}$$

and the analogous terminal-tail inclusions.

These reverse hit conditions are essential. Containment of the tail in a gate alone would allow an additional disconnected ball, or an uncontrolled corridor, from which no short internal secondary connector follows. We now verify the connectors supplied by the actual finite label.

Lemma 4.2 (Buffered plateau connections)**.** *Suppose the secondary internal source upper Hölder bounds hold in doubled balls throughout each buffered seed inspection, and the primary internal source upper bounds hold throughout the site inspection A, including every doubled ball used along its leads and retained tree. Every point of G_0 can be joined to $\gamma(0)$ inside G_0 at secondary source cost at most*

$$CH_E g^s + 2t_0\ell_1\Lambda_E. \tag{4.6}$$

Here Λ_E is the upper conversion from secondary own seed units to primary site units. The terminal assertion is identical. The whole plateau network has deterministic internal primary diameter

$$A_{\mathrm{net}} \le C(1+\mu^{-d}+N\omega^{-1}) \le Cv^{-C}. \tag{4.7}$$

No primary length is assigned to the secondary geodesic.

Proof. An internal upper bound in doubled balls implies that any point of a closed ball $\overline{B}_t(q)$ can be joined to q inside that ball at cost CHt^s. To include boundary points, use consecutive interior balls on a radius whose scales decrease geometrically to the boundary point. Their doubled balls stay inside $B_t(q)$, and the costs sum because $s>0$. The concatenation extends continuously to its limiting endpoint. Infima may be approximated with arbitrarily small total error.

For $x\in G_0$, choose $q\in Q_0$ with $x\in\overline{B}_g(q)$. The hit condition provides $\gamma(t_q)$, $t_q\le 2t_0$, within $c_d\mu$ of q. Join x to q inside that ball, then join q to $\gamma(t_q)$ in a small doubled ball still inside it. Follow the actual secondary tail to $\gamma(0)$; coverage in (4.2) keeps this tail in G_0. This proves (4.6). It also proves that every gate is path connected; no disconnected gate components are simply being grouped.

Give two marked full-arc cells an adjacency edge whenever their closed cells intersect, including an intersection only along an edge or corner. This adjacency graph is connected. Otherwise their separated finite closed unions would disconnect the connected arc while every component is hit. Adjacent centers are at distance $C_d\mu$, so primary upper Hölder connectors between them fit in radius-ω chamber balls. A spanning tree joins all centers, and the preceding radial construction attaches every chamber point. For a polygonal lead tube, sample its centerline at intervals at most $\omega/100$ and use internal connectors in its balls. There are at most $CN\omega^{-1}$ samples. The retained tree is treated in the same way. The rounded lead endpoints lie deep inside their gates; every gate center attaches to an actual arc point in its own gate ball. These connections give the deliberately coarse bound (4.7). □

Exact incidence. Thicken both the leads and every retained tree segment by ω, including their endpoint balls, and delete open gate interiors from the lead pieces, retained tree pieces, and chambers. Group pieces belonging to the same tree vertex into one vertex/lead remnant. All distinct vertex/lead and chamber remnants then have polynomial separation, including a vertex remnant and its own incident chamber. Only an intended gate–remnant pair may touch; all other gate pairs and gate–remnant pairs have polynomial separation. Here is the only nontrivial local check. If x is in the first lead tube outside $\operatorname{int} G_0$ and y is in the chamber outside that interior, choose an arc point within 2ω of y. If its time is at least t_0, lead clearance gives $|x-y|\ge r-3\omega$. Otherwise (4.5) gives $|x-y|\ge g-c_d\mu-2\omega$. Both estimates survive closure. The opposite lead is at distance at least $r-g-c_d\mu-\omega$ from G_0. The other endpoint is identical. The inverse arc modulus separates the gates, and distinct reserved boxes separate different edge constructions. Taking the minimum with the rational lead and port separations gives

$$\delta_{\mathrm{sep}} \ge c\nu^3 \ge cv^C. \tag{4.8}$$

Only a gate and its indicated vertex and chamber remnants may touch.

Here is an explicit order which verifies the polynomial assertion. Let $s > 0$ be a common source upper Hölder exponent. First take $t_0 = cv^p$ with $p > \max\{1, 1/(\alpha s)\}$, and $r = ct_0^{1/(\alpha\beta^2)}$. Choose the reciprocal rational mesh ν comparable to

$$c \min\{r^{3d+2}, t_0^{2/\beta}, v^{2/s}\},$$

with the port denominator as a divisor of its denominator. Then take $\omega = c\nu^4$ and $\mu = c\omega^2$, using comparable rational meshes where needed. This meets the candidate-count and clearance inequalities, and gives $A_{\mathrm{net}} \le C\nu^{-8d}$ and $\delta_{\mathrm{sep}} \ge c\nu^3$. Every choice depends only on the fixed source references and v, before any output geodesic is seen.

There are $O(\mu^{-d})$ cells and polynomially many lead vertices. Thus the complete finite catalogue satisfies

$$\log J \le C\mu^{-d} + CN \log \nu^{-1} \le Cv^{-C}. \tag{4.9}$$

For measurability, use the continuous intrinsic metric profiles on compact exhaustions of the finitely many buffered inspections, together with their finite unit-conversion ratios. Strong locality makes these Borel functions of the restricted field. The source arcs with the fixed two-sided moduli, diameter restriction and compact trial range form a compact path space. Include their own lengths in a fixed compact interval. Intrinsic geodesicity is the closed condition $E_A(\gamma(s), \gamma(t)) = \ell|s - t|$ at rational s, t. The favorable inequality, covered-and-hit labels, lead clearances and attachment constraints are closed as well. Projection over the compact path-and-length factor is Borel; strict inequalities are exhausted by positive rational gaps. Every favorable arc on the references has an admissible label. Selecting the first label does not divide its probability by the number of labels. Global pin caps used to establish physical unit conversion are external to the local block cover; the counting events may retain these Borel conversion constraints.

4.2 The local certificate

Normalize a site radius and its primary unit to one, and put

$$\mathcal{C} = \{3/2 \le |x| \le 5/2\}, \qquad A = \{1/2 < |x| < 7/2\}, \qquad B = B_4.$$

The local primary references give internal upper Hölder connectors, an intrinsic lower bound $H^{-1}\,|x - y|^b$, collector diameter $\mathrm{diam}_D(\mathcal{C}; A) \le A_0$, and positive lower bounds for both collar crossings. Choose $a_{\mathrm{out}} > 0$ below both collar bounds and below H^{-1}, before choosing any subsequent constant. Choose $b_{\mathrm{trig}} \ll a_{\mathrm{out}}$ and then $\varepsilon_0 \ll b_{\mathrm{trig}}$. A finite lattice tree in the interior of $\mathcal{C}$ gives source access cost at most ε_0 from every collector point. Choose its mesh so that each augmented vertex star, including the complete reserved boxes of all incident edges, has source diameter at most ε_0 using connectors inside A. Reserve disjoint convex boxes about its middle edges, with rational ports and fixed buffers. All these choices precede v.

On each edge require a short intrinsic secondary arc with fixed own length bounds and (4.1), satisfying, in primary site units,

$$a_e = D(u_e, v_e) = D(u_e, v_e; A_{\mathrm{seed}}) \in [d_0, D_0], \qquad \mathrm{len}_E\,\gamma_e \le (1 + 4v)a_e. \tag{4.10}$$

The equality with the global distance is a consequence of the local first-exit reference, valid for every exterior extension. Secondary internal upper bounds are converted using only a fixed upper bound on their own seed unit relative to the primary site unit. This conversion precedes modification: the seed pins themselves need not avoid the bump.

Lemma 4.3 (Finite site forcing)**.** *Under these local references, there is a finite deterministic catalogue of nonnegative smooth bumps f_ℓ and Borel source validity events, exhausting the favorable site event, such that*

$$\log J \le Cv^{-C}, \qquad \|f_\ell\|_{\mathrm{CM}}^2 \le C\xi^{-2}v^{-C}.$$

All bump supports lie strictly inside the collector and avoid the site pin. There are deterministic $\theta, c_{\mathrm{seed}} \ge cv^C$ with the following property. If $\tau\|f_\ell\|_\infty \le \Omega \ll v$, every modified primary geodesic with endpoints outside B which passes the source trigger

$$D(u_0, v_0; B) \ge b_{\mathrm{trig}}$$

at its first and last collector visits contains a block of modified primary length at least c_{seed}. The block has both endpoints in enlarged gates and lies wholly in A. Its endpoints admit an actual modified secondary path inside B with length at most $(1+6v)e^{C\Omega}$ times that block length. These statements are uniform under arbitrary exterior fields.

Proof. Use the preceding ball-cover construction on every edge. Let U be the union of its closed chambers and gates, its closed lead tubes, and the closed ω-tubes around all retained tree segments, including the vertex balls. This is the closed plateau network. The original straight middle edge segments are absent. Decrease t_0 and g so that both primary gate diameters and the secondary costs in (4.6) are at most $vd_0/1000$. Choose the deterministic discount from (4.7):

$$\theta A_{\mathrm{net}} \ll \varepsilon_0, \qquad \theta A_{\mathrm{net}} \ll H^{-1}\delta_{\mathrm{sep}}^b. \tag{4.11}$$

It can satisfy $\theta \ge cv^C$. Next choose a deterministic power scale h with

$$\begin{aligned} h \ll \delta_{\mathrm{sep}}, \qquad CHh^s \ll \min\{\varepsilon_0, H^{-1}\delta_{\mathrm{sep}}^b, vd_0\}, \\ CH_E h^s \le v\theta d_0/1000. \end{aligned} \tag{4.12}$$

For the preceding explicit geometry one may take

$$\theta = c\nu^{8d+3b+1}, \qquad h = c\min\{\nu^4, \nu^{(3b+1)/s}, (v\theta)^{2/s}\}.$$

Thus the order is geometry, discount, support collar; no target metric regularity is required to choose any of them. Set $V = U^{+h}$, and use G_a^{+h} as the itinerary gates. These enlarged gates are not assumed to lie in the plateau. Put

$$M = \xi^{-1}\log\theta^{-1}, \qquad f = M\mathbf{1}_{U^{+h/4}} * \varphi_{h/4}, \tag{4.13}$$

where φ is a nonnegative smooth unit-mass mollifier. Then $0 \le f \le M$, $f = M$ on U, and $\operatorname{supp} f \subset U^{+h/2}$, strictly inside V and the collector. The two collars and the site pin are unchanged. The primary Weyl weight is exactly $e^{-\xi f}$. The secondary weight differs from this common ideal weight by a factor between $e^{-\Omega}$ and e^{Ω}, since $\tau\|f\|_\infty \le \Omega$. Thus both metrics obey the bounds $[\theta e^{-\Omega}, e^{\Omega}]$ globally and $[\theta e^{-\Omega}, \theta e^{\Omega}]$ on U, with weight exactly one outside V. In fact the primary metric can only decrease. The Fourier multiplier bound for the smoothing mollifier gives, in every dimension,

$$\|f\|_{\dot H^{d/2}}^2 \le CM^2h^{-d}\,\|\mathbf{1}_{U^{+h/4}}\|_2^2 \le C\xi^{-2}\log^2\theta^{-1}h^{-d} \le C\xi^{-2}v^{-C}.$$

This is the Cameron–Martin norm up to its fixed covariance constant. It is invariant under site dilation and unaffected by subtracting the pinning constant. In particular, no boundary-area estimate for the labeled sets enters the energy.

Let P be a modified primary geodesic as in the statement. Source access and the discounted network give a competitor between its first and last collector visits of cost at most $3\varepsilon_0$. Any excursion of this interval outside A crosses an unchanged collar twice and costs at least $2a_{\mathrm{out}} > 3\varepsilon_0$. Its entire image is therefore in A.

Any two points of V have a modified network competitor of cost

$$B_h = e^{\Omega}(2CHh^s + \theta A_{\mathrm{net}}).$$

Choose (4.11)–(4.12) so that $B_h < H^{-1}(\delta_{\mathrm{sep}}/2)^b$. An excursion outside the closed set V whose endpoints in V are at least $\delta_{\mathrm{sep}}/2$ apart stays in A and is unchanged, since the actual support lies strictly inside V. Its source length is at least $H^{-1}(\delta_{\mathrm{sep}}/2)^b$ by the intrinsic lower bound. The network competitor would shorten the geodesic, so such an excursion is impossible.

Write K_v for the closed vertex/lead remnants and K_e for the closed chamber remnants after gate interiors are removed. Suppose no chamber's opposite enlarged gates are both visited. In the subdivided incidence tree

$$K_v \;-\; G_{v,e} \;-\; K_e \;-\; G_{w,e} \;-\; K_w,$$

delete every gate node whose h-enlargement is not visited, retaining all remnant nodes. For each remaining graph component take the union of the h-neighborhoods of its node sets. Distinct such unions have distance at least $\delta_{\mathrm{sep}} - 2h \geq \delta_{\mathrm{sep}}/2$: all distinct remnant pairs and gate pairs, and all nonincident gate–remnant pairs, have separation δ_{sep}; an intended incidence between retained nodes would put them in the same component. Every visit to V lies in one of these unions, since the deleted enlarged gates are not visited. If two unions were visited, start with the union containing the first V-point. Take the first visit to another union and the last visit to the initial union before it. These hitting times exist because the finitely many unions are closed. The whole intervening open interval is outside V, giving an excursion with endpoint separation at least $\delta_{\mathrm{sep}}/2$, a contradiction. Thus all visits to V, if there are any, lie in one remaining graph component.

Such a component contains at most one original tree vertex: passage between two original vertices would require both opposite gate nodes on an intervening edge. Its node sets therefore lie in that vertex's augmented star. A component with no original vertex is an isolated chamber, which belongs to either endpoint's augmented star. Hence, whenever V is visited, its first and last points lie within h of one augmented vertex star. Disconnected remnants only enlarge this graph description. No finiteness assertion about the number of visits or regularity constant of the modified geodesic is used.

The interval must visit V, since otherwise it is unchanged and has source cost below the trigger. Its initial and terminal tails outside V each have source cost at most $3\varepsilon_0$. In the single-star alternative, two source h-attachments and the star connector would give

$$D(u_0, v_0; B) \leq 7\varepsilon_0 + 2CHh^s < b_{\mathrm{trig}},$$

a contradiction. Hence opposite gates of some chamber are visited. The intervening block lies in A and has modified primary length

$$L \geq \theta e^{-\Omega}\{a_e - \mathrm{diam}_D(G_0^{+h}) - \mathrm{diam}_D(G_1^{+h})\} \geq \theta d_0/2 =: c_{\mathrm{seed}}. \tag{4.14}$$

For each block endpoint in G_a^{+h}, take a source secondary connector to G_a in a small ball of the seed inspection. Its cost $CH_E h^s$ is charged at full density, up to e^{Ω}, using (4.12). The connector from there to u_e or v_e lies completely in the plateau gate by Lemma 4.2 and is discounted. Follow the actual favorable arc between them in its plateau chamber. Comparing this cost with (4.14),

and using (4.10) with the stated error margins, gives $(1+6v)e^{C\Omega}L$. This actual secondary shortcut stays in the inspection. The primary block need not follow the arc or remain in its chamber. All arguments occurred before any exterior field could be encountered. □

Two additional consequences will be used in the count. A source geodesic visiting the inner half-ball, with both endpoints outside B, has an actual subcrossing from the outer to the inner boundary of $\mathcal{C}$, entirely within that collector. Its endpoints are Euclidean distance at least one apart, so the intrinsic lower reference bounds its source length below by $H^{-1} \geq a_{\mathrm{out}}$, by the initial choice of a_{out}. Its network replacement saves a fixed $a_0 > 0$ before the small Weyl-mismatch allowance. Source subcrossings at disjoint inspections have disjoint time intervals. Also, removing a discount whose current trigger fails costs at most b_{trig} in its source site unit: replace its entire first-to-last collector interval by a source connector inside B. If the collector is not visited, the removal costs zero. Other selected supports are outside this inspection, and retained portions avoid the removed support. Finally, arbitrary first-to-last intervals can be restored chronologically, from the first remaining hit through its last visit; the chosen restoration intervals are disjoint and each costs at most A_0 in its source site unit.

4.3 One common output and one source comparison

Lemma 4.4 (Finite Gaussian overlap)**.** *Let G have a Gaussian law, and let deterministic Cameron–Martin shifts f_i, $i \in \mathcal{I}$, satisfy $\|f_i\|_{\mathrm{CM}} \leq M$, where $|\mathcal{I}| = N < \infty$. If measurable events satisfy*

$$\sum_i \mathbf{1}_{A_i}(G) \geq L\mathbf{1}_B(G), \qquad \sum_i \mathbf{1}_{A_i}(G+f_i) \leq U$$

almost surely, then

$$\mathbb{P}(B) \leq \frac{U}{L}\min\{N, e^{M\sqrt{2\log N}}\}. \tag{4.15}$$

No independence among events or shifts is required.

Proof. Let $W(f)$ be the Gaussian linear functional with variance $\|f\|^2_{\mathrm{CM}}$. Cameron–Martin change of variables gives

$$L\mathbb{P}(B) \leq \mathbb{E}\sum_i \mathbf{1}_{A_i}(G+f_i)e^{-W(f_i)-\|f_i\|^2_{\mathrm{CM}}/2} \leq U\mathbb{E}\max_i R_i,$$

where $R_i = e^{-W(f_i)-\|f_i\|^2_{\mathrm{CM}}/2}$. Since $\mathbb{E}R_i = 1$, the last expectation is at most N. For $p > 1$ it is also at most $N^{1/p}\exp((p-1)M^2/2)$. If $M > 0$ and $N > 1$, set $p = 1 + \sqrt{2\log N}/M$ and use $1/p \leq 1/(p-1)$. This gives the other bound. The remaining cases are immediate. □

Here is the precise probabilistic interface. Common reference and site-cover failures are outside this proposition; they must be paid once before a later endpoint net is introduced.

Proposition 4.5 (Forcing count for a fixed endpoint pair)**.** *Fix the local reference geometry of Lemma 4.3, $C_b > 1$, $c \in [1+20v, C_b]$, and a prescribed $\mu > 0$. Consider two fixed separated endpoints $x, y \in B_2$. Work under $h_{10}(0) = 0$. A deterministic radius/color/unit-bin catalogue $\mathfrak{C}$ has at most $C\Delta^{-D_0}$ possible sites, with disjoint full inspections (the closed radius-four balls in site coordinates), all possible supports lie in B_4, and both endpoints lie outside every possible full inspection. Its deterministic primary unit bin is $[Q, 2Q]$, where $Q \geq \Delta^{K_0}$. The radius-ten field pin is outside all possible supports. At a source field H, let $B_{\mathfrak{C}}$ require the conditions below. A selection*

of catalogue bumps always means at most one labeled bump per physical site; only such selections are used in the proposition. Assume every candidate bump satisfies $\tau\|f_{z,\ell}\|_\infty \le \Omega$*, where* Ω *obeys* (4.17) *below.*

(i) $E_H \le cD_H + \eta_{\mathrm{loc}}$ *for every endpoint pair in a fixed guarded ball* $B_{R_{\mathrm{src}}}$*;*

(ii) $E_H(x,y) \ge (c - g_\Delta)D_H(x,y)$ *and* $D_H(x,y) \le U_0$*;*

(iii) an internal source competitor of cost at most U_0 *and an unchanged surrounding annular barrier of cost greater than* $2U_0$*, confining the primary geodesic after every legal negative selection of catalogue bumps to* $B_{R_{\mathrm{src}}}$*;*

(iv) at least $N_{\mathrm{site}} \ge \Delta^{-1/2}$ *eligible sites of this catalogue, with actual source units in* $[Q,2Q]$*, visited in their inner cores by the canonical source primary geodesic.*

For deterministic constants sufficiently small, take

$$g_\Delta = c_g v c_{\mathrm{seed}} \Delta^{K_0}/U_0, \qquad \eta_{\mathrm{loc}} \le c_\eta v c_{\mathrm{seed}} \Delta^{K_0}, \tag{4.16}$$

$$\Omega \le c_\Omega \min\{v, a_0\Delta^{K_0}/U_0, v c_{\mathrm{seed}}\Delta^{K_0}/(C_b U_0)\}. \tag{4.17}$$

If $\xi \ge a > 0$ *and* $\log \Delta^{-1}$ *exceeds a fixed polynomial in* a^{-1}, v^{-1}*, then*

$$\mathbb{P}(B_{\mathfrak{C}}) \le \Delta^{2\mu+2}.$$

The polynomial can depend on μ *and the fixed reference geometry, but not on an outer scale, outer output accuracy, or outer failure probability. A logarithmic number of catalogues can consequently be summed with total bound* Δ^μ*; the same holds for any fixed power of* $1 + \log \Delta^{-1}$*.*

Proof. All primary site units are unchanged under every catalogue bump: the site pins lie in their central holes, and the inspections are disjoint. The global radius-ten pin also avoids every support. Fix one valid label at every initially supplied site. Let k be chosen below and $m_0 = \lceil \kappa k \rceil$, where $\kappa > 0$ is a fixed constant smaller than $a_0/(8A_0)$. Index all labeled subsets Z of possible sites with $m_0 \le |Z| \le k$, and write $f_Z = \sum_{i\in Z} f_{i,\ell_i}$. Define $A_Z(H)$ by $B_{\mathfrak{C}}(H)$, validity of every selected source label, including all its fixed local references and unit-conversion bounds, actual membership of every selected source unit in $[Q,2Q]$, and passage of every selected trigger by the canonical geodesic of $H - f_Z$. The selected sites need not belong to the initial supplied set in this definition. This is why endpoint exclusion was imposed on the entire deterministic catalogue. Actual supports at different sites are disjoint; hence a bound Ω on each bump's Weyl mismatch also bounds the error of their sum. One may instead allocate Ω/k to each bump.

Lower multiplicity. Choose any k of the N_{site} supplied sites. Their genuine source collector subcrossings are disjoint in time. Replace them by the discounted networks. The exact primary discount cannot increase the remaining source path portions; the weaker factor e^Ω is harmless. The source length bound and (4.17) therefore leave a saving at least $7a_0kQ/8$. Delete the least-index trigger-failing site, recompute the canonical modified geodesic, and repeat. Each deletion costs at most $2b_{\mathrm{trig}}Q$, and an unvisited collector costs zero. Choose $b_{\mathrm{trig}} \le a_0/32$. At most k deletions occur and a saving at least $a_0kQ/2$ remains. All predicates and the finite pruning rule are Borel.

For the surviving set Z, restore chronologically: take the first remaining selected collector hit, replace through its last visit by a source connector, and continue after that interval. Retained pieces avoid every selected support; inserted source connectors lie in disjoint inspections. Hence

$$D_H(x,y) \le D_{H-f_Z}(x,y) + 2A_0\,|Z|\,Q.$$

It follows that $|Z| \geq \kappa k$, and $A_Z(H)$ holds. A fixed surviving labeled set has at most $N_{\mathrm{site}}^{k-m_0}$ original k-subsets as preimages. Provided $N_{\mathrm{site}} \geq k$,

$$\sum_Z \mathbf{1}_{A_Z}(H) \geq N_{\mathrm{site}}^{m_0}/k^k \quad \text{on } B_{\mathfrak{C}}. \tag{4.18}$$

No supply estimate along intermediate modified geodesics was used.

One output geodesic. Every selected bump is already pinned: its support lies in B_4, whereas the Gaussian law is pinned on the sphere of radius ten. Thus the Cameron–Martin shift is literally f_Z. At output h, an active index has source $H = h + f_Z$ and modified field exactly h. Every active index therefore has the same canonical primary output geodesic P_h. All primary site units are unchanged, since their sphere-one pins lie in the central holes of their inspections and the inspections are disjoint. There is no repinning by a common scalar, which would be invalid for unequal parameters.

Let $\mathcal{V}$ be the union of sites in active indices. At every site i choose one active certificate $Z \ni i$, write $f_i = f_{i,\ell_i}$, and use the single-site source $H_i^* = h + f_i$. Removing the other disjoint shifts from $h + f_Z$ preserves its local metric profiles, label validity, internal trigger, and seed and site pins: both the other bump supports and their metric modification supports lie outside the full inspection of i. The source global endpoint equality for its accepted arc also persists, since it follows from the local first-exit bound for every exterior extension. As $H_i^* - f_i = h$, Lemma 4.3 applies directly to P_h and supplies a block of length at least $c_{\mathrm{seed}}Q$ and its local secondary shortcut. In fact the complete first-to-last collector intervals J_i, $i \in \mathcal{V}$, are pairwise disjoint. To see this, for a fixed i use one active source containing it. Its network gives a competitor between those visits costing at most $3\varepsilon_0 S_i$. Its source collars are exactly unchanged, since all other selected supports are outside its inspection. Leaving its annular inspection would cost two collar crossings, contradicting geodesicity. Thus $P_h[J_i]$ is contained in that inspection. Distinct inspections are disjoint, so the intervals are disjoint as well. This argument uses each site's source only for its local references; no global comparison at those other sources is invoked.

One source comparison. Choose one active Z_0 and put $H_0 = h + f_{Z_0}$. Discard the blocks at its at most k sites. Keep one block at every site of $\mathcal{V} \setminus Z_0$, so $m = |\mathcal{V} \setminus Z_0| \geq |\mathcal{V}| - k$. Their intervals and complete secondary shortcuts are disjoint from selected inspections and remain unchanged at H_0. Restore the selected collector intervals by their H_0-source connectors, costing at most $2A_0Q$ each. Use the m shortcuts and apply the comparison at H_0 to each residual interval and restoration jump. Retained primary portions avoid all selected collectors, so their h- and H_0-lengths agree exactly. Their endpoints lie in the comparison domain by the source barrier. Target connectors may leave that domain because the comparison concerns global distances. There are at most $m + k + 1$ residual intervals and k restoration jumps. Consequently, with L_i the retained primary block lengths,

$$\begin{aligned} E_{H_0}(x,y) \leq cD_h(x,y) - [c - (1+6v)e^{C\Omega}] \sum_{i=1}^{m} L_i \\ + 2cA_0kQ + (m + 2k + 1)\eta_{\mathrm{loc}}. \end{aligned} \tag{4.19}$$

Only this one source comparison has been used.

Since $D_h \leq e^{\Omega} D_{H_0}$, $D_{H_0}(x,y) \leq U_0$, and $c \geq 1 + 20v$, the negative term saves at least $12vc_{\mathrm{seed}}mQ$, after reducing the constant in (4.17). Compare (4.19) with the near-bad source

inequality. The errors (4.16)–(4.17) absorb the term linear in m and give

$$|\mathcal{V}| \le K_1(v)k, \qquad K_1(v) \le C\{1 + A_0/(vc_{\mathrm{seed}})\}.$$

Hence the number of active labeled subsets is at most

$$\sum_Z \mathbf{1}_{A_Z}(h + f_Z) \le (CK_1J)^k. \tag{4.20}$$

Gaussian summation and depth. There are at most $(CJ\Delta^{-D_0})^k$ deterministic indices. The triangle inequality in Cameron–Martin space gives $\|f_Z\|_{\mathrm{CM}} \le kM_0$, with $M_0^2 \le Ca^{-2}v^{-C}$. Apply Lemma 4.4 to (4.18) and (4.20). Writing $\ell = \log \Delta^{-1}$ and $\rho = 1/2$, we obtain

$$\begin{aligned}\log \mathbb{P}(B_{\mathfrak{C}}) \le & -\rho\kappa k\ell + k\log k + k\log(CK_1J) \\ & + kM_0\sqrt{2k\{D_0\ell + \log(CJ)\}}.\end{aligned} \tag{4.21}$$

Choose $k \ge (4\mu + 4)/(\rho\kappa)$, then impose

$$\ell \ge \max\left\{\frac{\log(CJ)}{D_0}, \quad \frac{4\{\log k + \log(CK_1J)\}}{\rho\kappa}, \quad \frac{64kD_0M_0^2}{\rho^2\kappa^2}, \quad \rho^{-1}\log k\right\}.$$

The positive terms in (4.21) each consume at most a quarter of its negative term. Thus its right side is at most $-(2\mu + 2)\ell$. The depth requirement is polynomial in a^{-1}, v^{-1} by (4.9) and the energy bound. It involves none of the outer output parameters. Any fixed power of the logarithm in the number of catalogues is absorbed by the remaining power of Δ. □

4.4 From the count to the local improvement

Proof of Proposition 3.5. All fixed probabilities below are chosen small enough in terms of the p_0 in Proposition 3.3. Work under $h_{10}(0) = 0$. We first choose the source geometry, before a, v. Take an upper quantile $U_0/4$ for a compact internal competitor simultaneously for pairs in B_8, and a positive lower quantile for one unit distance. The relative escape estimate, with a sufficiently large fixed factor, gives an annular barrier outside B_8 whose cost exceeds $4U_0$, with arbitrarily small fixed failure. Fix $R_{\mathrm{src}} \ge 16$ containing that barrier and competitor, and then fix the global primary Hölder reference on this actual ball. Choose $R_s \ge R_{\mathrm{src}}$. All catalogue supports will lie in B_4. The barrier is therefore unchanged by every negative catalogue modification, and its strict lower bound exceeds the required absolute threshold $2U_0$.

Lemma 3.4 supplies a favorable own-unit arc with fixed positive probability in each seed inspection. For each edge of the fixed collector tree take a fixed number of line-separated seed trials. Equation (2.9) makes the probability that all trials fail at most $(1 - p_*/2)^{N_{\mathrm{tr}}/2}$. Choose N_{tr} first, then a sufficiently small fixed seed-to-site radius ratio to fit the trials inside their reserved boxes. After these choices, bound the finitely many seed-to-site pin differences in fixed intervals, losing an arbitrarily small fixed probability. They are local coordinates in the whole site inspection. The local references for both metrics and these pin differences give a fixed upper conversion of secondary seed units into secondary site units. This is an own-site local statement.

Put $\ell = \log \Delta^{-1}$, to be chosen as a polynomial in a^{-1}, v^{-1}. Use all separated site radii in $[\Delta^2, \Delta]$, with their own-radius center grids in B_3. The whole-site event just constructed is a local event in separate own normalizations and has as high a fixed probability as needed for (2.10). Separately, the Gaussian cap for the finite site and seed pin lists gives

$$|\log(S_\eta/S_\xi)| \le C\tau\ell \tag{4.22}$$

at every relevant seed and site, outside $C\Delta^c$. Require $C\tau\ell \ll v$. Multiplying the preceding seed-to-site conversion by the capped ratio S_η/S_ξ at the site now bounds secondary seed units in primary site units. At each seed the same cap turns the favorable ratio $1+2v$ in own units into at most $1+4v$ in physical units. The site event together with this separate cap therefore supplies a label satisfying (4.10). The cap in (4.22) is paid outside the block cover. The probability of a failed simultaneous site cover is at most $C\Delta^c$.

On the fixed global primary reference the primary unit of each eligible site obeys, for one fixed K_0,

$$\Delta^{K_0} \le S_\xi(s,z) \le \Delta^{-K_0}. \tag{4.23}$$

For the lower bound use two collector points at separation of order s: their global lower distance is at least $H^{-1}s^\beta$, whereas their local competitor costs at most $A_0 S_\xi(s,z)$. For the upper bound every path from the center to distance $3s$ pays the collector crossing $a_{\mathrm{out}}S_\xi(s,z)$, while the global upper bound supplies a competitor of cost $H(3s)^\sigma$. Both bounds imply (4.23), after increasing K_0.

For any source pair in B_2 separated by at least $w_*/2$, first stop its primary geodesic when it reaches radius $w_*/4$ about the first endpoint. Use the segment from its last visit to radius $w_*/8$ before that stopping time through the stopping time. Core visits at radial-level gaps 10Δ give at least c/Δ disjoint inspections, avoiding both endpoints and lying in B_4. Partition them by radius, a fixed center-grid residue color, and the dyadic primary bins in (4.23). There are only a fixed power of $1+\ell$ such catalogues, so one has at least $\Delta^{-1/2}$ supplied sites. Restrict every candidate catalogue, not only the supplied subset, to inspections avoiding both endpoints. Its cardinality is polynomial in Δ^{-1}, and all its supports are inside the radius-ten pin.

Let $\mathcal{R}$ contain the fixed primary reference, the barrier and compact competitor, the site cover and the separate pin cap. Its failure is at most a prescribed small p_{com} when ℓ is large. It holds simultaneously for all relevant endpoint pairs. Let G be the source comparison in the premise of (3.13). For a fixed separated deterministic pair, apply Proposition 4.5 on $G\cap\mathcal{R}$. Choose

$$g_\Delta = c\,v c_{\mathrm{seed}}\Delta^{K_0}/U_0, \qquad \varepsilon_* \le c\,v c_{\mathrm{seed}}\Delta^{K_0}. \tag{4.24}$$

The unequal Weyl allowance is valid for every label if

$$\tau\, C a^{-1}\log v^{-1} \le c\min\{v, a_0\Delta^{K_0}/U_0, v c_{\mathrm{seed}}\Delta^{K_0}/(C_b U_0)\}. \tag{4.25}$$

This costs only $\tau \le \exp[-P(a^{-1},v^{-1})]$. The count and the sum over catalogues bound the probability of $E(x,y)\ge (c-g_\Delta)D(x,y)$ on $G\cap\mathcal{R}$ by Δ^μ.

To make this simultaneous, round endpoints on a deterministic net in $\overline{B}_2$ of mesh t_Δ, chosen so that

$$CC_bHt_\Delta^\sigma + 4\varepsilon_* \le g_\Delta d_{\mathrm{sep}}/100. \tag{4.26}$$

Here $d_{\mathrm{sep}}>0$ is the fixed primary lower bound for pairs separated by $w_*/2$. Primary increments use H; secondary increments use the source comparison on G. A pair separated by at least $3w_*/4$ and violating coefficient $c-g_\Delta/8$ rounds to a near-bad net pair at coefficient $c-g_\Delta$. The pair net has cardinality at most $\Delta^{-D_{\mathrm{net}}}$ for a fixed exponent: all powers of v^{-1} are absorbed by $\ell \ge C\log v^{-1}$. Fix $\mu > D_{\mathrm{net}}+2$ before choosing the polynomial value of ℓ. The resulting probability of some separated bad pair is at most

$$q_0 + p_{\mathrm{com}} + C\Delta^{\mu-D_{\mathrm{net}}}. \tag{4.27}$$

The common failures q_0, p_{com} are paid once, outside the net.

On the fixed local references, a witness of $H_\uparrow$ gives one of these separated global bad pairs by the subdivision in Section 2. Both the count and $H_\uparrow$ use the radius-ten pin, so this passage entails no change of comparison coefficient. Set $g = g_\Delta/16$. Choose q_0, p_{com} and the local-reference failure sufficiently small, and then choose $\ell = P_1(a^{-1}, v^{-1})$ sufficiently large. Equation (4.27) proves $\mathbb{P}(H_\uparrow(c-g)) \le p_0/2$. The count requires a polynomial lower bound on ℓ; (4.24) and (4.26) require only a polynomial upper bound on $\log \varepsilon_*^{-1}$. Choose one such ε_* and then impose (4.22) and (4.25). All these choices are independent of c, and yield exactly (3.12). □

5 The common continuous family

The increment estimate concerns deterministic parameter pairs. We now use finite parameter grids to obtain one event for the entire family. The final nondegeneracy argument is needed because positivity at each deterministic parameter alone would not exclude a random exceptional parameter.

5.1 Positive parameter intervals

Fix a compact interval $I = [a, b] \Subset (0, \xi_c)$ with rational endpoints, and an integer R. Set $X(\xi) = \log(1 + \rho_h^\xi)$ as a random element of $C(\overline{B}_R^2)$. Let

$$\delta_j = 2^{-j-6}, \qquad k_j = \lceil A 2^{Dj} \rceil, \qquad G_j = \{a + (b-a)m2^{-k_j} : 0 \le m \le 2^{k_j}\}, \tag{5.1}$$

where $D > B$ and then A is sufficiently large in terms of I, R and Proposition 3.7. The grids are nested; let $w_j = (b-a)2^{-k_j}$ be their spacing and take $n_j = c_1 \delta_j / w_j$, with a sufficiently small fixed c_1. These choices satisfy (3.17) for every pair at distance at most w_j.

Compare each node of G_{j+1} with both bracketing nodes of G_j, and compare adjacent nodes of G_j. There are at most $C 2^{k_{j+1}}$ such pairs. Their total failure at tolerance δ_j is at most

$$C\, 2^{k_{j+1}} \delta_j^{-M} \exp\{-c\delta_j^{M+1}/w_j\}. \tag{5.2}$$

This is summable: its positive logarithmic terms are $O(2^{D(j+1)} + j)$, whereas its negative term is a fixed multiple of $2^{k_j - (M+1)j}$. Borel–Cantelli therefore makes all these comparisons hold for sufficiently large j, on one event.

Linearly interpolate the Banach-valued profiles at G_j. The bracketing comparisons bound the difference of two successive interpolants by $2\delta_j$ uniformly on I. They converge uniformly to a continuous profile X_I. The function $\rho_I = e^{X_I} - 1$ is continuous in the parameter and endpoints. On the dense union of grids it agrees with the original metrics, so symmetry, the diagonal condition and triangle inequality pass to every parameter.

For each deterministic $\xi \in I$, this is a version of the prescribed metric. Indeed, (3.16) implies stochastic continuity: for fixed tolerance choose $n = c\delta/|\xi - \eta|$ and then let $\eta \to \xi$. The seed condition eventually holds and the failure tends to zero. Grid values converging to ξ converge almost surely to ρ_I^ξ and in probability to the prescribed ρ^ξ. The limits agree almost surely.

Carry this out on countably many rational intervals and integer balls. The versions agree on overlaps: they agree almost surely at a countable dense set of deterministic parameters, and both are continuous. Thus they give one field-measurable, locally uniformly continuous family of finite pseudometrics for $0 < \xi < \xi_c$.

5.2 The Euclidean endpoint

Only a simple marginal estimate from the supplied first-order paper is needed here.

Lemma 5.1 (Marginal Euclidean bound)**.** *For every $R, p < \infty$ and $r < 1$, for all sufficiently small deterministic $\xi > 0$,*

$$\left\| \|\rho_h^\xi - |\cdot - \cdot|\|_R \right\|_{L^p} \le C_{R,p,r}\xi^r. \tag{5.3}$$

The assertion holds under either fixed sphere pin.

Proof. The estimates in [9, (3.20)–(3.21), Proposition 3.2 and the display preceding (7.4)] give

$$\sup_{x,y\in B_R} \|\rho_h^\xi(x,y) - |x-y|\|_{L^q} \le C_{R,q}\xi \tag{5.4}$$

and every prescribed moment of a spatial Hölder constant of exponent $\beta_0 < 1$, uniformly for sufficiently small ξ. The supremum in (5.4) is outside the norm. Choose $r < r_0 < \beta_0 < 1$, a spatial mesh of spacing $\ell = \xi^{r_0/\beta_0}$, and a fixed $q \ge p$ so large that $1 - 2dr_0/(\beta_0 q) \ge r_0$. The maximum over the $O(\ell^{-2d})$ grid pairs has L^p norm at most $C\xi\ell^{-2d/q} \le C\xi^{r_0}$; interpolation adds at most $C\ell^{\beta_0} = C\xi^{r_0}$. This proves (5.3). The radius-ten pin differs by the factor in (2.5); Gaussian exponential moments and Hölder's inequality make its additional L^p error $O(\xi)$. □

We prove the endpoint on the already constructed positive-parameter version. In particular, continuity on a positive interval will extend bounds on a dense grid to that whole interval. Let a be dyadic and small, and set $\varepsilon_a = \sqrt{a}$. For any deterministic pair in $[a, 2a]$, Lemma 5.1 with $r = 3/4$ gives the fixed-confidence seed at accuracy $c\varepsilon_a$. Hence the last sentence of Proposition 3.7 applies at tolerance $c'\varepsilon_a$ to every such pair, with no generic seed condition. Its remaining depth and drift conditions are polynomial in a^{-1}.

Here are finite grids that prevent any exceptional parameter spikes. Choose $D > B(1 + 1/2) + M + 2$ and A large, using the fixed constants in the generic increment estimate. The generic fine grids on $[a, 2a]$ have relative dyadic levels

$$k_j(a) = \lceil Aa^{-D}2^{Dj}\rceil, \quad j \ge 0, \qquad \delta_j(a) = 2^{-j-6}\varepsilon_a. \tag{5.5}$$

Their spacings $w_j = a2^{-k_j(a)}$ satisfy the generic seed condition at accuracy $\delta_j(a)$. With $n_j = c\delta_j(a)/w_j$, the total failure of all successive bracketing comparisons is bounded by the analogue of (5.2). Uniformly for small a, its sum over j is smaller than every fixed power of a.

It remains to reach $G_0(a)$ without paying its exponential cardinality using a marginal bound. Choose

$$n_0^{\mathrm{base}} = a^{-L}, \qquad w_{\mathrm{base}} \asymp c\varepsilon_a/n_0^{\mathrm{base}}, \qquad L > M/2 + D + 3, \tag{5.6}$$

rounding the base spacing downward to a relative dyadic spacing. There are only polynomially many base nodes. Apply (5.3) and Markov's inequality to make all their profiles within ε_a of $X_0 = \log(1 + |\cdot - \cdot|)$, outside probability Ca^m for any prescribed m. The moment order is chosen after L, D; shrinking the eventual range of a does not change those already fixed exponents.

Now compare every node of the fine grid $G_0(a)$ with a bracketing base node, at tolerance $\varepsilon_a/8$ and depth n_0^{base}. This uses the cheap seed above. The union bound is at most

$$C\varepsilon_a^{-M}\exp\{CAa^{-D} - c\varepsilon_a^M a^{-L}\} \le \exp\{-c'a^{-L+M/2}\}. \tag{5.7}$$

The inequality follows from the last choice in (5.6). There is only this one preliminary bridge. The generic fine comparisons then bound every node in their dense union within $2\varepsilon_a$ of X_0, after

increasing a fixed constant. Positive-parameter continuity extends this to all of $[a, 2a]$. Taking $m = 2$ and summing over dyadic a, Borel–Cantelli proves

$$\lim_{\xi \downarrow 0} \|\rho_h^\xi - |\cdot - \cdot|\|_R = 0 \quad \text{almost surely.} \tag{5.8}$$

The passage from logarithmic profiles uses that X_0 is bounded on B_R^2 and the displayed errors tend to zero. Intersect over integer R and set $\rho_h^0 = |\cdot - \cdot|$.

5.3 No exceptional degenerate parameter

Choose a countable norm-dense rational subgroup $\mathcal{H}_0$ of the Cameron–Martin space consisting of bounded pinned smooth shifts. For completeness, covariance potentials of smooth compactly supported zero-mass tests are bounded and smooth; subtract their radius-ten pin. A countable dense set of such tests and its rational span gives $\mathcal{H}_0$. Intersect the constructed good event with all of its translates by $\mathcal{H}_0$. Cameron–Martin equivalence leaves an invariant event of probability one. Fixed-parameter Weyl scaling on a countable dense parameter set, followed by continuity, gives there

$$e^{-\xi\|f\|_\infty} \rho_h^\xi \le \rho_{h+f}^\xi \le e^{\xi\|f\|_\infty} \rho_h^\xi, \qquad f \in \mathcal{H}_0, \quad 0 < \xi < \xi_c. \tag{5.9}$$

For a compact positive interval I and disjoint compact rational balls A, B with positive separation, put

$$Z(h) = \{\xi \in I : \rho_h^\xi(A, B) = 0\}. \tag{5.10}$$

It is a random compact set. Extend it by the empty set off the invariant good event. Equation (5.9) makes Z invariant under every shift in $\mathcal{H}_0$. For every relatively open parameter interval U, the event $\{Z \cap U \neq \varnothing\}$ is measurable: exhaust U by compact sets and minimize the continuous nonnegative function $\xi \mapsto \rho_h^\xi(A, B)$.

Each such event has probability zero or one. Indeed, if its indicator is Y and $p = \mathbb{E}Y$, invariance and Cameron–Martin change of variables give

$$\mathbb{E}\Big[Y e^{W(f) - \|f\|_{\mathrm{CM}}^2/2}\Big] = p, \qquad f \in \mathcal{H}_0. \tag{5.11}$$

Continuity of these exponentials in L^2 extends the identity to all Cameron–Martin vectors. Their totality in finite-dimensional Gaussian L^2, followed by conditioning on increasing finite coordinate sets, gives $Y = p$ almost surely. The hitting indicators of a countable basis determine a closed set, so Z is deterministic almost surely. For every deterministic parameter the prescribed metric is nondegenerate, and the constructed family is its version. A nonempty deterministic Z would contain a deterministic parameter contradicting this fact. Thus Z is empty. Countably many intervals and rational-ball pairs show that every member of the family is a genuine metric on one event.

5.4 The length property and the median

Only countably many field dilations are needed to obtain common properness. Let $Th = h(2\,\cdot) - h_{20}(0)$ under the radius-ten pin. This preserves the pinned Gaussian law. Intersect the preceding good event with all integer iterates of T. Exact strong covariance holds there first at countably many dense parameters and then, by continuity, at every parameter. For $I \Subset (0, \xi_c)$ let

$$W(h) = \min_{\xi \in I} \rho_h^\xi(\partial B_1, \partial B_2) > 0. \tag{5.12}$$

Poincaré recurrence applied to $\{W \geq 1/k\}$ gives, almost surely, a random k and arbitrarily large j such that $W(T^j h) \geq 1/k$. Gaussian tails give $h_{10\,2^j}(0)/j \to 0$. Consequently the exact scaling identity implies along those j,

$$\min_{\xi \in I} \rho_h^\xi(\partial B_{2^j}, \partial B_{2^{j+1}}) \geq k^{-1} 2^{j \min_I \phi} e^{-(\sup I)|h_{10\,2^j}(0)|} \longrightarrow \infty. \tag{5.13}$$

The first-exit inequality holds at dense parameters and passes to every parameter by continuity. Thus (5.13) confines metric balls uniformly over I. Continuity and nondegeneracy identify the Euclidean and metric topologies on each compact, so the metrics are proper and induce the Euclidean topology globally.

For an arbitrary parameter, approximate it by parameters in the dense set and take their canonical geodesics between fixed endpoints. The common barriers confine these geodesics to one compact. Parametrize at constant metric speed. Positive separation of distinct points on that compact and uniform convergence of the metric profiles give Euclidean equicontinuity. Arzelà–Ascoli and the geodesic equality then give a geodesic for the limiting parameter. This argument is deterministic on the common good event and applies to every parameter and endpoint pair.

Return now to the unit-sphere-pinned field using (2.5). The factor $e^{\xi h_{10}(0)}$ is a continuous function of ξ, equal to one at zero, so all the continuity conclusions persist. Let

$$m_\xi = \operatorname{med} \rho_h^\xi(0, e_1), \qquad m_0 = 1. \tag{5.14}$$

The supplied characterization gives a unique unit-distance median [8, Lemma 10.3]. Weak continuity of the laws gives continuity of m_ξ: for every $t > 0$, uniqueness implies $\mathbb{P}\{\rho_h^\xi(0, e_1) \leq m_\xi - t\} < 1/2$ and $\mathbb{P}\{\rho_h^\xi(0, e_1) \geq m_\xi + t\} < 1/2$. The two closed-half-line Portmanteau inequalities force all nearby medians into $(m_\xi - t, m_\xi + t)$. The marginal Euclidean bound gives $m_\xi \to 1$ at zero. Thus m_ξ is positive and continuous on $[0, \xi_c)$. For each deterministic ξ, the metric $m_\xi^{-1} \rho_h^\xi$ is the unique median-one metric. Its common version is continuous by what we have proved. Finally use the parameter homeomorphism (1.5). This proves Theorem 1.1. □

A Uniform one-metric estimates on subcritical parameter intervals

We prove the parameter-uniform form of the one-metric estimates used in the comparison argument. The proof uses the finite-range approximation of Ding–Gwynne–Zhuang [1]; it does not assume continuity or uniqueness of any limiting metric. Throughout this appendix, $B_r(x)$ is a Euclidean ball and $\mathcal{Q}_r(x) = x + (-r, r)^d$ is an axis-aligned cube.

Fix $d \geq 3$ and one nonnegative, radial, smooth, compactly supported kernel K with $\int K^2 = 1$. With space-time white noise W, set

$$h_{m,n}(x) = \int_{2^{-n}}^{2^{-m}} \int_{\mathbb{R}^d} K\left(\frac{x - y}{t}\right) t^{-(d+1)/2}\, W(dy, dt), \qquad h_n = h_{0,n}.$$

Let L_n^ξ be the exponential length metric with density $e^{\xi h_n}$, and use a semicolon for internal distances. Write

$$a_n^\xi(q) = \inf\{a > 0 : \mathbb{P}[L_n^\xi(0, e_1; \mathcal{Q}_2) \leq a] \geq q\}, \qquad a_0^\xi(q) = 1.$$

The choice of $\geq q$ rather than $> q$ is immaterial at positive ξ; at zero every quantile is one. Put $\phi(\xi) = \xi Q(\xi)$ for $\xi > 0$ and $\phi(0) = 1$. Fix a compact interval $I \Subset [0, \xi_c)$ and write

$$b = \sup I, \qquad \mu = \inf_{\xi \in I} \{\phi(\xi) - \sqrt{2d}\,\xi\} > 0. \tag{A.1}$$

Continuity at zero, and hence the positivity in (A.1), also follow from the argument below. All constants may depend on I, K, d and fixed inspection shapes. No constant depends on the cutoff or on $\xi \in I$.

A.1 A uniform version of the DGZ quantile package

Proposition A.1 (Uniform quantiles and tightness)**.** *For every $q \in (0,1)$ and $\vartheta > 0$, there is $C < \infty$ such that*

$$C^{-1} 2^{(1-\phi(\xi)-\vartheta)m} \leq \frac{a^{\xi}_{n-m}(q)}{a^{\xi}_{n}(q)} \leq C 2^{(1-\phi(\xi)+\vartheta)m}, \qquad 0 \leq m \leq n. \tag{A.2}$$

For every connected open U and compact $E \subset U$, the internal metrics $L^{\xi}_n(\cdot,\cdot;U)/a^{\xi}_n(q)$ are uniformly tight on $E \times E$. For each $0 < s < \mu$, their Euclidean s-Hölder seminorm on E is tight. For every fixed cubical shell $\mathcal{A}$ of positive width,

$$\lim_{u \downarrow 0} \sup_{n, \xi \in I} \mathbb{P}\Big[L^{\xi}_n(\text{across } \mathcal{A}) < u a^{\xi}_n(q)\Big] = 0. \tag{A.3}$$

Fixed buffered internal diameters have uniformly tight upper tails in the same units. All fixed quantiles are comparable:

$$C^{-1}_{p,q} \leq a^{\xi}_n(p)/a^{\xi}_n(q) \leq C_{p,q}, \qquad p, q \in (0,1). \tag{A.4}$$

Every subsequential limit along a deterministic array (ξ_j, n_j) of the normalized global metrics is a metric inducing the Euclidean topology.

Proof. We give the uniformity argument in dependency order; all references in the proof are to [1].

Uniform exponential orders. Let F^{ξ}_n be a unit distance, buffered diameter, or fixed-shell crossing. All its paths lie in a fixed box V, and

$$|\log F^{\xi}_n - \log F^{\eta}_n| \leq |\xi - \eta| \sup_V |h_n|. \tag{A.5}$$

Lemma 2.3 gives $\mathbb{E} \sup_V |h_n| \leq C(n+1)$. Thus $\xi \mapsto (n \log 2)^{-1} \mathbb{E} \log F^{\xi}_n$ has a common Lipschitz constant. Propositions 3.1, 4.7 and 4.8, together with Lemmas 2.7 and 4.6, identify its pointwise limit as $\phi(\xi) - 1$. At zero this limit is zero. For the unit functional this proves that ϕ extends continuously to zero. A finite parameter net then upgrades each pointwise limit to a uniform limit on I. Gaussian concentration, whose variance proxy is at most $b^2 n \log 2$, transfers the conclusion to every fixed log-quantile:

$$\sup_{\xi \in I} \left| \frac{\log \mathrm{Quant}_q(F^{\xi}_n)}{n \log 2} - (\phi(\xi) - 1) \right| \longrightarrow 0. \tag{A.6}$$

The same argument applies to $\sup_C \mathrm{diam}_{L^{\xi}_n}(\mathcal{Q}_1 \cap C; \mathcal{Q}_2 \cap C)$, where C ranges over cubes of side at least 10 containing zero: its pointwise exponent is the clipped version in footnote 11, and (A.5)

survives the supremum. Its log is a finite Lipschitz functional of the field on $\mathcal{Q}_2$, so measurability causes no additional issue. The extra buffered endpoint shapes in footnote 13 are covered by Lemma 3.6 and the same argument.

Uniform concentration and cutoff comparison. Choose $0 < \eta < \mu$ and $Q_0 > \sqrt{2d}$ so that $\phi(\xi) - \xi Q_0 > \eta$ throughout I. The coarse good-field event in Lemma 4.2 therefore fails with probability Ce^{-cm} uniformly. Fine good vertices and the two-level nice vertices have fixed dependence range and bad probability at most $C_d(1-q)$ by their own quantile definitions. Lemma 2.9 yields a common good-quantile threshold in Lemma 4.2(1). Its second claim is unnecessary here. In particular, no multiplicative quantile gap is imposed at $\xi = 0$.

For Lemmas 4.9–4.10, use their coarse depth $m = \lfloor n/(\log n)^2 \rfloor$. The coarse field exceeds an ηn allowance with probability at most $Ce^{-n\log n}$. By (A.6) and Gaussian concentration, a fine cell's diameter or crossing predicate fails with probability Ce^{-cn}, uniformly. Finite-range percolation bounds the relevant bad-cluster failure by $C2^{dm}e^{-c2^m} \le Ce^{-n\log n}$. For the around distance use the clipped predicate of footnote 11, so the detour stays in the prescribed containing cube. All remaining factors are $2^{O(m)}$ and fit inside the exponent allowance.

Choose the constants for Section 5 as follows. Put $Q_* := \inf_{I\cap(0,\infty)} Q > \sqrt{2d}$, and choose

$$0 < \alpha < \beta - d, \quad d + \alpha < \beta < Q_*^2/2, \quad \lambda = \frac{\alpha}{4(\alpha+d)}, \quad \sqrt{2\beta} < \bar{Q} < Q_*. \tag{A.7}$$

Put $\nu = \inf_I(\phi - \xi\bar{Q}) > 0$. Denote the coarse-graining integers of Section 5 by K, L, and their levels by $\ell_1 = \mathsf{K}^4$, $\ell_{j+1} = \lfloor (1+\lambda)\ell_j \rfloor + 1$. Lemma 5.5 is uniform because its base box has finitely many cells, its gradient has a fixed Gaussian law, and its remaining predicates follow from (A.6) with a fixed slack. In Lemma 5.6, write $u = \ell_{j-1}$ and $v = \ell_j$. Its sufficient coarse-field variable H has the law $\sup_{\mathcal{Q}_{4R_0 2^{-(v-u)}}} h_{v-u}$, where R_0 is the fixed geometric constant in Section 5. Its domain has bounded size in cutoff units. The sufficient inequality, before division by ξ, is

$$\xi H < \phi(\xi)(v-u)\log 2 - \theta(v+u)\log 2 - \mathsf{K}^{3/2}\sqrt{v}. \tag{A.8}$$

Since $\lambda u \le v - u \le \lambda u + 1$ and $u \ge \mathsf{K}^4$, on $H \le \bar{Q}\lambda u \log 2$ the margin is at least $\nu\lambda u \log 2 - \theta(3+\lambda)u\log 2 - C\mathsf{K}^{-1/2}u$. First choose θ small, then K large. Lemma 2.3(5) gives a failure exponent with leading coefficient $(\bar{Q}^2/2)\lambda u \log 2 > \beta\lambda u \log 2$; the two fine distance failures are superexponentially smaller. There is no spatial entropy term proportional to $v - u$ for this cutoff-cell supremum. This proves the uniform Lemma 5.6 even at zero.

Keep the actual detour predicate in Definition 5.3, rather than replacing it by the sufficient event used in (A.8). Its independence from the central bad-box sigma-algebra gives the same recursion as Lemma 5.7:

$$q_j \le C\big(e^{-\mathsf{K}\ell_j} + 2^{2d\lambda\ell_{j-1}} q_{j-1}^2 + 2^{(d-\beta)\lambda\ell_{j-1}} q_{j-1}\big). \tag{A.9}$$

The induction and top-box union use respectively $(2d+\alpha)\lambda < \alpha$, $\beta - d > \alpha$, and $\lambda < \alpha/[2(\alpha+d)]$, all consequences of (A.7). Choose L after K so that $q_1 \le 2^{-\alpha\mathsf{K}^4}$. Lemmas 5.8–5.9 now have uniform constants. Impose also $\mathsf{K} > (16db+1)^2$. In Proposition 5.10, the two coarse-field transfers lose at most $16d\xi\mathsf{K}\sqrt{v}$. The detour advantage $\mathsf{K}^{3/2}\sqrt{v}$ therefore leaves at least $\mathsf{K}\sqrt{u}$, uniformly for $\xi \le b$. In Lemma 5.11 the shell overlap is at most 100^d, and its detour cost is absorbed by choosing K so that

$$2\,100^d \sum_j e^{-\mathsf{K}\sqrt{\ell_j}} < 1. \tag{A.10}$$

The cell connectors in Lemma 5.12 have factors at most $\mathsf{L}\exp\{C_d b(\mathsf{L}+1)\}$, since their scaled coarse gradient is at most $\mathsf{L}+1$. The boundary connectors of footnote 13 require only the finitely many

fixed clipped shapes already covered by Lemma 3.6 and (A.6). Composing these constructions in Proposition 5.13 thus introduces only constants depending on the parameters already fixed above.

For clarity, Lemma 5.14 also has a uniform repair in its required range $0 \le k \le cn$, with $c \le 1$ decreased below if necessary. Its endpoint diameter sum starts at a level at least $\lambda n - O(1)$, and $n + k \le 2n$. Choose its chaining exponent to be a fixed fraction of μ, and choose the error in (A.6) much smaller than $\lambda\mu$. The endpoint diameters are then bounded by $2^{(\phi-1-u_0)(n+k)}$ for one $u_0 > 0$, outside probability $C2^{-cn}$. At $l = \lfloor \delta n \rfloor$, require the coarse field on each relevant cell to be at least $-Al$, with A large enough that its failure exponent beats $dl \log 2$. Use a fine crossing exponent error $v_0 < u_0/4$. The logarithmic margin over the endpoint threshold is at least

$$(u_0 - v_0)(n + k) \log 2 - (\sup_I \phi \log 2 + bA)l > 0$$

for a sufficiently small fixed δ. The union over $O(2^{dl})$ cells retains exponentially high probability. This proves Proposition 5.1 uniformly without dividing by ξ.

Before same-depth quantiles are comparable, its all-depth consequence is the spare-window statement: for $0 < p < q < r < 1$,

$$C^{-1}2^{(1-\phi-\vartheta)m} a_n^\xi(p) \le a_{n-m}^\xi(q) \le C2^{(1-\phi+\vartheta)m} a_n^\xi(r). \tag{A.11}$$

For $m \le cn$, use the cutoff comparison and put its probability loss inside the fixed window. For $m > cn$, use (A.6) with error smaller than $c\vartheta/4$. Bounded depths follow from the raw finite-cutoff field bound.

Upper tightness, a crossing witness, and escape. For $0 < s < \eta < \mu$ and $\vartheta < \eta - s$, Proposition 4.5 and (A.11) give, on the successful scales starting at r,

$$\sup_{|x-y|_\infty \le 2^{-k}} L_n^\xi(x, y; \mathcal{Q}_2) \le C a_n^\xi(q_0) \sum_{m=k}^{n} m^{2d} 2^{-(\eta-\vartheta)m} \le C a_n^\xi(q_0) 2^{-sk}, \quad r \le k \le n, \tag{A.12}$$

for one fixed $q_0 < 1$. The event has probability $1 - Ce^{-cr}$. Below the cutoff, use the smooth density as in (6.2); larger distances use a finite grid of these connections. Finite buffered covers give Lemma 6.2 and Proposition 6.3 with one admissible $q_2 < 1$. For deleted coarse scales on an unbounded domain, first choose bounded buffered competitors; the coarse supremum is taken only on that bounded set. This stage has used no lower crossing estimate.

Fix $q > q_2$. The FKG proof of Lemma 6.5 is uniform using $\mathbb{P}[L_n^\xi(0, e_1; \mathcal{Q}_2) < a_n^\xi(q)] \le q$. For a sufficiently small fixed shell, the endpoint zigzag uses a bounded number J of translated and rotated copies inside $\mathcal{Q}_2$. Positive association bounds each small endpoint-distance event by $q^{1/J} < 1$. Halve its threshold to include possible atoms. Let $c_0, p_0 > 0$ be the resulting uniform point-pair witness constants in a fixed containing box. Choose a fixed neighborhood radius $\delta > 0$, smaller than its boundary buffer, so that each boundary-centered neighborhood has internal diameter greater than $c_0 a_n^\xi(q)/4$ with probability at most $p_0/4$. The triangle inequality gives probability at least $p_0/2$ for each increasing event that the internal distance between an inner-boundary neighborhood and an outer-boundary neighborhood exceeds $c_0 a_n^\xi(q)/2$. Take finite covers of both boundaries by these neighborhoods and apply FKG to all these neighborhood-separation events. Their intersection forces the shell crossing above that threshold and has uniformly positive probability, as in Lemma 6.6. No modulus failure is subtracted after this finite intersection.

For escape, the scaling comparison for a radius-2^k shell is

$$L_n^\xi(\text{across } \mathcal{Q}_{2^{k+1}} \setminus \mathcal{Q}_{2^k}) \succeq 2^k e^{-\xi \sup_{\mathcal{Q}_2} h_k} L_{n+k}^\xi(\text{across } \mathcal{Q}_2 \setminus \mathcal{Q}_1).$$

Use the witness at a fixed $q' > q$, (A.11), and $\sup h_k \le (\sqrt{2d}\log 2 + \delta)k$. With fixed positive probability the right side exceeds $ca_n^{\xi}(q)e^{[(\mu-\vartheta)\log 2 - b\delta]k}$. Take ϑ, δ small. Sufficiently separated large shells are independent for the stationary finite-range field. Summing their crossings proves the uniform version of Lemma 6.7.

Uniform nondegeneracy. A finite grid on a shell's inner boundary reduces its crossing lower bound to that of small concentric cube annuli: every shell crossing contains one of those grid annular crossings. By translation invariance, it therefore suffices to treat $\mathcal{Q}_{2r} \setminus \overline{\mathcal{Q}_r}$ for fixed $0 < r < 1/10$. If (A.3) fails there, finite-depth bounds give a sequence $n_j \to \infty$, $\xi_j \in I$ with crossing mass uniformly bounded away from zero at thresholds tending to zero. Extract jointly all normalized internal metrics on bounded connected finite unions of rational open boxes, for every fixed number m of deleted scales, and also all normalized global metrics and coarse fields h_m on integer boxes. This countable domain family includes the disjoint neighborhoods of compact zero sets used below; single cubes would not suffice. Upper tightness and Skorohod give almost sure local uniform convergence. On each bounded domain U,

$$e^{-b\sup_U |h_m|} L^{\xi_j}_{m,n_j}(\cdot,\cdot;U) \le L^{\xi_j}_{0,n_j}(\cdot,\cdot;U) \le e^{b\sup_U |h_m|} L^{\xi_j}_{m,n_j}(\cdot,\cdot;U).$$

Because the coarse fields were included in the coupling, the limits have identical zero sets with and without deleting m scales. On two positively separated domains, the deleted-scale internal metrics are independent once m is large; their limits are independent as well. This argument needs bounded ξ_j, not parameter continuity of a limiting metric.

We record the boundary point in the zero-set argument of Lemma 6.9. For a bounded domain U, let D be the global limit, D_U an internal limit, and $S \Subset U$ a reference sphere. The global zero neighborhood $E = \{x : D(x,S) = 0\}$ is compact and connected by escape and the Hausdorff-limit argument of Lemma 6.8. Let K_U be the closure in $\overline{U}$ of $\{x \in U : D_U(x,S) = 0\}$, and let $\widehat{E}_U$ be the connected component of the compact set K_U containing S. These are measurable compact sets: the zero-set closure is Borel from compact-set minima, and its component is the intersection of the sets reachable from S by finite Euclidean $1/j$-chains in K_U. This compact-set encoding is Borel. By $D_U \ge D$ and global continuity, $K_U \subset E$. Then

$$\widehat{E}_U \Subset U \implies D(\partial U, S) > 0 \implies \widehat{E}_U = E. \tag{A.13}$$

For the first implication, paths of length tending to zero from S to ∂U are stopped first on compact inner exhaustions of U. Hausdorff limits lie in the internal zero component and reach every inner boundary, contradicting its compact containment. Only convergence on compact subsets of U is used. For the second, first-exit paths give $D(x,S) \ge \min\{D_U(x,S), D(\partial U, S)\}$ in U, and $D(x,S) \ge D(\partial U, S)$ outside; also $D_U \ge D$.

Here is the resulting zero–one argument explicitly. Put $S_1 = \partial\mathcal{Q}_r$, $S_2 = \partial\mathcal{Q}_{2r}$, $E_i = \{x : D(x,S_i) = 0\}$ and $p = \mathbb{P}[E_1 \cap E_2 = \varnothing]$. By the triangle inequality and compactness this equals $\mathbb{P}[D(S_1,S_2) > 0]$. For a compact connected set F, let $\Gamma_j(F)$ be the union of the open cubes $\mathcal{Q}_{2^{-j}}(z)$ with $z \in 2^{-j}\mathbb{Z}^d$ and $\operatorname{dist}_\infty(z,F) < 2^{1-j}$. This is a bounded connected domain in our countable rational family; it contains F and lies within $3 \cdot 2^{-j}$ of it in the ℓ^∞ distance. Write $A_{i,j}(U) = \{\Gamma_j(E_i) = U\}$. For a candidate U containing S_i, (A.13) and $\widehat{E}_{i,U} \subset E_i$ give the event identity

$$A_{i,j}(U) = \{\Gamma_j(\widehat{E}_{i,U}) = U\}.$$

Indeed either equality puts the compact internal component inside U, whereupon it equals E_i. Thus, for each fixed pair U, V with positive separation, deletion of sufficiently many fixed coarse

scales proves $\mathbb{P}[A_{1,j}(U) \cap A_{2,j}(V)] = \mathbb{P}[A_{1,j}(U)]\mathbb{P}[A_{2,j}(V)]$. No random domain is conditioned upon. Summing these equalities gives

$$\begin{aligned} p &= \lim_{j\to\infty} \sum_{\operatorname{dist}_\infty(U,V)>2^{-j}} \mathbb{P}[A_{1,j}(U)]\mathbb{P}[A_{2,j}(V)] \\ &\le \lim_{j\to\infty} \left(\sum_{U\cap S_2=\varnothing} \mathbb{P}[A_{1,j}(U)] \right) \left(\sum_{V\cap S_1=\varnothing} \mathbb{P}[A_{2,j}(V)] \right) = p^2. \end{aligned}$$

The first limit follows because two disjoint compact zero neighborhoods have positive separation; the marginal sums tend to p because $E_1 \cap S_2 = \varnothing$ and $E_2 \cap S_1 = \varnothing$ are each equivalent to $E_1 \cap E_2 = \varnothing$. The positive witness gives $p > 0$, whereas the assumed array gives $p < 1$: the boundary-pair infimum is continuous under local uniform convergence. This contradiction proves uniform nondegeneracy. Fixed finite covers extend it to every prescribed shell of positive width.

Finally, the unit distance dominates a fixed crossing. The lower estimate and upper tightness compare every fixed unit quantile with one high fixed quantile, proving (A.4). Substitution in (A.11) gives (A.2). Countably many compact crossings rule out zero distances in every global subsequential limit; upper continuity and escape give its Euclidean topology, as in the proof of Theorem 1.2. □

Remark A.2 (The clipped exponent can also be obtained directly). On the ordinary successful good-component event, put $\delta_m = Am^2 2^{-m}$. For $x \in \mathcal{Q}_1 \cap C$, clamp each coordinate into the interval at distance δ_m from the two faces of C, and choose a nearby vertex in the common good component at levels $m-1, m$. Consecutive vertices have distance at least $\frac{1}{2}(A - C_0)m^2 2^{-m}$ from every clipping face. Choose A larger than the tube constant in Lemma 4.1. Their level-m connector and all its doubled-box inspections stay in C; a fixed large starting level keeps them in $\mathcal{Q}_2$. This works for every C on one event. The terminal straight attachment stays in the convex intersection. Ordinary unit exponential orders then bound the sum by $C2^{(\phi-1+\vartheta)n}$, and the clipped supremum dominates a unit distance. Thus this construction independently supplies the exponent invoked from footnote 11, without a union over clipping faces.

A.2 Moments, confined moduli, and relative escape

Let κ be the CHZ convolution kernel associated with the fixed K, and let h be the whole-space LGF with unit-sphere average zero. Define $L_{\xi,\varepsilon}$ from $h * \kappa_\varepsilon$, and normalize by $b_{\xi,\varepsilon} = \exp \mathbb{E} \log L_{\xi,\varepsilon}(0, e_1)$, setting $D_{\xi,\varepsilon} = L_{\xi,\varepsilon}/b_{\xi,\varepsilon}$. The logarithmic integrability implicit in this definition is proved below, before escape or geodesic existence is used.

Proposition A.3 (The uniform metric inputs). *For each compact $I \Subset [0, \xi_c)$ there exist $p_* > 0$, $p_{\mathrm{esc}} > 0$, $0 < \sigma < 1 < \beta$, and finite constants C, A with the following properties. For every deterministic $\xi \in I$, $0 < \varepsilon \le 1$, and $R \ge 1$, there is a random $H_R \ge 1$ such that*

$$\sup_{\xi\in I,\, 0<\varepsilon\le 1} \mathbb{E} H_R^{p_*} \le C(1+R)^A, \qquad \begin{cases} \operatorname{diam}_{D_{\xi,\varepsilon}}(B_s(z); B_{2s}(z)) \le H_R s^\sigma, & z \in B_R,\ 0 < s \le 1, \\ D_{\xi,\varepsilon}(x,y) \ge H_R^{-1}|x-y|^\beta, & x, y \in B_R. \end{cases} \tag{A.14}$$

Every fixed compact shell crossing has uniformly bounded inverse moments of all finite orders. Fixed compact internal diameters have a positive moment, and fixed nonzero point distances have all finite logarithmic moments. Moreover,

$$C^{-1} r^{\phi(\xi)+\vartheta} \le r\, b_{\xi,\varepsilon/r}/b_{\xi,\varepsilon} \le C r^{\phi(\xi)-\vartheta}, \qquad \varepsilon \le r \le 1, \tag{A.15}$$

for every fixed $\vartheta > 0$. Each fixed smooth metric is proper and geodesic, and for deterministic $R, S \geq 1$,

$$\mathbb{P}[\text{a geodesic with endpoints in } B_R \text{ leaves } B_{SR}] \leq CS^{-p_{\mathrm{esc}}}. \tag{A.16}$$

The expectation in (A.14) *is taken before the supremum over parameters and cutoffs; it is a uniform marginal bound.*

Proof. We first derive the two moment inputs for the stationary approximation, using Proposition A.1.

Confined upper chaining. Choose $\bar{Q} > \sqrt{2d}$, $\vartheta > 0$ and $s_0 \in (0,1)$ with $\phi(\xi) - \xi\bar{Q} - \vartheta \geq s_0 > 0$ on I. Gaussian mesh-point tails, followed by the cutoff-cell oscillation bound, give

$$\sup_{B_{R+10}} h_m \leq \bar{Q}m \log 2 + C\log(2+R) + Z_R, \qquad \mathbb{P}[Z_R > u] \leq Ce^{-cu}, \quad m \geq 0. \tag{A.17}$$

Indeed, an intermediate slope in $(\sqrt{2d}, \bar{Q})$ beats the $O((1+R)^d 2^{dm})$ mesh points; the remaining slope controls cell oscillations, and $C\log(2+R)$ absorbs spatial volume.

Declare fine vertices good when all needed neighbor connectors are at most $C_q 2^{-m} a^{\xi}_{n-m}$. Their bad probability can be arbitrarily small by uniform upper tightness. Wire the labels good outside a fixed padded box, and compute the adjacent-scale nice predicate from these already wired fields. Finite-range coloring and lattice animals give

$$\mathbb{P}[M_R > s] \leq C(1+R)^d e^{-cs^2}, \tag{A.18}$$

where M_R is one plus the last scale below n with a bad star cluster of diameter larger than Cm^2, including the adjacent-scale predicate. Let $\mathcal{O}_m$ be the unique infinite good cluster in the wired level-m field. On successful scales, every point is within $Cm^2 2^{-m}$ of a vertex in $\mathcal{O}_{m-1} \cap \mathcal{O}_m$: the infinite wired nice cluster is contained in both clusters, exactly as in the proof of DGZ (4.14). Two vertices of $\mathcal{O}_m$ at separation ℓ have a good path in a $Cm^2 2^{-m}$-tube around their segment, with at most $Cm^{2d}(1 + 2^m \ell)$ edges. The vertices chosen in every endpoint chain belong to these infinite clusters; arbitrary good vertices could instead lie in finite enclosed islands. This is the finite wired geometry of Lemma 4.1: fill the finite holes of each encountered bad star cluster and bypass its exterior good layer, then erase loops. All paths used below lie inside the padded box, so exterior wired edges are never used as metric connectors.

By (A.2) and (A.17), the full cost of a good edge, divided by a^{ξ}_n, is at most

$$C(2+R)^C e^{bZ_R} 2^{-s_0 m}. \tag{A.19}$$

For $2^{-j-1} < s \leq 2^{-j}$, start at a successful scale

$$m = \max\{j, M_R, m_*\} + O(\log(j + M_R + 2)) \quad \text{with } Cm^2 2^{-m} < s/100.$$

Connect the two initial good vertices, then use common vertices at finer levels for each endpoint chain. Their tubes and all doubled-cell inspections lie in $B_{2s}(z)$. Summing (A.19), for any $0 < \sigma < s_0$, gives

$$\operatorname{diam}_{L^{\xi}_n / a^{\xi}_n}(B_s(z); B_{2s}(z)) \leq C(2+R)^C e^{CM_R + bZ_R} s^{\sigma}. \tag{A.20}$$

The strict gap $s_0 - \sigma$ absorbs all powers of the level and its logarithmic excess over j; when $M_R > j$, the loss is e^{CM_R}. The last vertex is $O(n^2 2^{-n})$ from its endpoint, so its straight attachment costs at most $C(2+R)^C e^{bZ_R} n^2 2^{-s_0 n}$ in normalized units. If the selected starting scale is at least n, use a straight segment throughout. Its cost is at most $C(2+R)^C e^{bZ_R} s 2^{(1-s_0)n}$, and

$n - j \le M_R + O(\log(j + M_R + 2))$ gives the same bound. These observations cover all radii below the cutoff too.

Take $H_R^+ = C(2+R)^C e^{CM_R + bZ_R}$. Equation (A.18) gives every finite exponential moment of M_R, with polynomial growth in R. Choose $p_+ > 0$ with $2p_+ b < c$ in (A.17) and apply Cauchy–Schwarz. This proves a uniform positive moment of H_R^+ and of compact internal diameters.

All inverse crossing moments. Use (A.3) to choose a fine-cell lower crossing quantile c_0 with bad probability below the finite-range lattice-animal threshold. Except with probability Ce^{-c2^m}, every macroscopic crossing visits a good inner cell. Indeed, otherwise its lattice trace contains a bad star chain of diameter $c2^m$. Choose m so a doubled cell has diameter less than the separation of the macroscopic endpoints. At a good visit, at least one endpoint lies outside the doubled cell; the intervening subpath completes that cell's annular crossing.

On the coarse event $\xi \inf h_m \ge -Am \log 2$, the ratio estimate therefore gives, for the normalized fixed-shell crossing Y_n,

$$Y_n \ge c2^{-Bm}, \qquad B = A + \sup_I \phi + \vartheta + 1.$$

The coarse failure is at most $C\exp[-c(A - C_0 b)^2 m/b^2]$. For L large take $m = \lfloor L/(2B\log 2)\rfloor$. If $m \le n$, then

$$\mathbb{P}[Y_n < e^{-L}] \le Ce^{-c\exp(cL/B)} + Ce^{-c(A-C_0b)^2 L/(b^2 B)}. \tag{A.21}$$

If $m > n$, the raw minimum density and $a_n \le Ce^{Cn}$ show that this event requires a negative Gaussian excursion of order L/b; its probability is at most $Ce^{-cL^2/(b^2 n)} \le Ce^{-cBL/b^2}$. Both exponential rates tend to infinity with A. Given any finite $q > 0$, take A large enough that they exceed $2q$ and integrate the tail. At $b = 0$ the assertion is deterministic. Thus

$$\sup_{n,\xi\in I} \mathbb{E}Y_n^{-q} < \infty \qquad \text{for every finite } q > 0. \tag{A.22}$$

Transfer and normalization. We recall the Gaussian facts from the convolution setup used in this step. At each fixed cutoff, the convolution and full white-noise fields agree in law after subtracting their own sphere averages [2, Proposition 1.10]. Their pin corrections are centered with uniformly bounded Gaussian exponential moments. On B_R, the remaining infrared correction has supremum exponential moments growing at most polynomially in $1 + R$. A real cutoff is reduced to an integer cutoff by removing a top band of width at most one and rescaling by a factor in $[1, 2]$. These are field comparisons; none identifies the processes jointly over cutoffs. Hölder's inequality transfers (A.20) at a smaller positive order and (A.22) at every prescribed finite order.

Every raw global unit distance is sandwiched between a mandatory compact crossing and a compact internal competitor. Its logarithm in a_n units is therefore uniformly integrable, and $b_{\xi,\varepsilon} \asymp a^{\xi}_{\lfloor \log_2 \varepsilon^{-1}\rfloor}$. The centered own-pin corrections give equality, rather than an extra deterministic shift, of the two geometric-mean normalizers. The integer floors differ by at most one, proving (A.15). The same sandwich gives all finite logarithmic moments of point distances.

Global lower modulus. Set $\Phi \ge \max\{1, \sup_I \phi + \vartheta\}$. For the normalized radius-r shell crossing $Z_\varepsilon(z, r)$, exact pinned scaling, (A.15), inverse moments and Gaussian pin variance give

$$\mathbb{E}Z_\varepsilon(z,r)^{-q} \le C_q(1 + |z|)^{Cq^2b^2} r^{-q\Phi - Cq^2b^2}, \qquad 0 < r \le 1. \tag{A.23}$$

For $r \ge \varepsilon$, use Cauchy–Schwarz for the rescaled field and its pin. For $r < \varepsilon$, the raw lower bound is $cr\varepsilon^{\Phi-1}e^{\xi \inf_{B_{Cr}(z)} h_\varepsilon}$. The cutoff-cell oscillation has bounded Gaussian moments, while the center variance is $O(\log \varepsilon^{-1} + \log(2+|z|))$. The resulting inverse bound is $C_q(1+|z|)^{Cq^2b^2} r^{-q}\varepsilon^{-q(\Phi-1)-Cq^2b^2}$; because $\Phi \ge 1$, it is bounded by (A.23). Thus no moment estimate at a cutoff larger than one is used.

There are at most $C(1+R)^d r^{-d}$ grid shells at a dyadic radius r in B_{R+3}. With $q=1$, choose $\beta > \Phi + Cb^2 + d + 2$. Markov's inequality and the convergent radius sum give a $C(1+R)^A u^{-1}$ tail for $\sup_{z,r} r^\beta / Z_\varepsilon(z,r)$, hence every moment below one. Every path between separated endpoints contains one of these tested crossings, even if it later leaves the observation box. This proves the lower bound in (A.14); separations larger than one cost only an additional power of $1+R$. Combine its constant with the upper constant at a common smaller positive moment order.

Properness and relative escape. Write $c_{\xi,\varepsilon}(r) = r b_{\xi,\varepsilon/r}/b_{\xi,\varepsilon}$ for the deterministic factor in the exact scaling of the normalized smooth metric. Choose ϑ so $p_0 := \inf_I \phi - \vartheta > 0$. Inverting (A.15) gives $c_{\xi,\varepsilon}(S) \geq C^{-1} S^{p_0}$ for $S \geq 1$. For the normalized crossing $A_\varepsilon(S)$ between radii $S/2$ and S,

$$\mathbb{E} A_\varepsilon(S)^{-q} \leq C S^{-qp_0 + Cq^2 b^2}. \tag{A.24}$$

Choose a small $q > 0$ so the exponent is at most $-qp_0/2$. A compact internal diameter V_ε is one competitor for all endpoint pairs in B_1 and has a positive moment p_+. Splitting at $S^{p_0/4}$ gives

$$\mathbb{P}[A_\varepsilon(S) \leq 2V_\varepsilon] \leq C S^{-p_0 \min(q,p_+)/4}. \tag{A.25}$$

On dyadic S, (A.24) with a threshold tending to infinity proves divergent barriers almost surely for each smooth metric. Thus it is proper; compactness for length metrics gives geodesics. Equation (A.25) confines every geodesic with endpoints in B_1 on one event. For B_R, scale both the barrier and competitor by R. Their common factor $c_{\xi,\varepsilon}(R) e^{\xi h_R(0)}$ cancels, and the transformed cutoff $\varepsilon/R \leq 1$ belongs to the same class. This proves (A.16) with $p_{\mathrm{esc}} = p_0 \min(q,p_+)/4$. □

Corollary A.4 (Preselected high orders near zero)**.** *For every finite $p, M > 0$ and $0 < \sigma < 1 < \beta$, there is $b_0 > 0$ such that Proposition A.3 holds on $[0, b_0]$ with $\mathbb{E} H_R^p \leq C(1+R)^A$ and escape exponent at least M. The field, kernel and normalization remain the same as in the compact subcritical construction; only the parameter range and constants change.*

Proof. Keep a fixed $\bar{Q} > \sqrt{2d}$. Since $\phi(\xi) \to 1$, choose the ratio slack small and then shrink b_0 so the upper chaining exponent s_0 exceeds the prescribed σ. The bad-scale variable has every finite exponential moment; the coarse multiplier requires only $2p' b_0 < c$ for a finite enlargement p' of the desired moment order. Choose this order before shrinking b_0, also allowing for Hölder transfer to the full field. For the lower modulus choose $q > p$ with $q(\beta - 1) > d + 2$, then shrink the slack and b_0 until $q(\beta - \Phi) - Cq^2 b_0^2 > d + 1$ in the dyadic sum following (A.23). Finally choose finite q, p_+ large enough in (A.25) to obtain M, and shrink b_0 again until $Cq^2 b_0^2 \leq qp_0/2$. Here p_0 stays bounded away from zero. All choices involve only a finite list of moment orders. □

References

[1] Jian Ding, Ewain Gwynne and Zijie Zhuang. *Tightness of exponential metrics for log-correlated Gaussian fields in arbitrary dimension.* arXiv:2310.03996v5, 24 July 2026. https://arxiv.org/abs/2310.03996v5.

[2] Andres A. Contreras Hip and Zijie Zhuang. *Weak exponential metrics for high-dimensional log-correlated Gaussian fields.* arXiv:2512.06292v1, 6 December 2025. https://arxiv.org/abs/2512.06292v1.

[3] Andres A. Contreras Hip and Zijie Zhuang. *Bounds on the distance exponent for higher-dimensional Liouville first passage percolation.* arXiv:2504.09141v1, 12 April 2025. https://arxiv.org/abs/2504.09141v1.

[4] Jian Ding and Ewain Gwynne. *Uniqueness of the critical and supercritical Liouville quantum gravity metrics*. arXiv:2110.00177v3, 2 September 2022. Section 4, the field-modification counting argument. https://arxiv.org/abs/2110.00177v3.

[5] Wei-Kuo Chen, Nikos Dafnis and Grigoris Paouris. *Improved Hölder and reverse Hölder inequalities for correlated Gaussian random vectors*. arXiv:1306.2410v2, 7 October 2013. https://arxiv.org/abs/1306.2410v2.

[6] Allen Hatcher. *Algebraic Topology*. Cambridge University Press, 2002. Section 3.3, Alexander duality and its naturality. https://pi.math.cornell.edu/~hatcher/AT/AT.pdf.

[7] Konstantinos Kavvadias. *Continuity in ξ and $\xi \to \infty$ tightness of the LQG metric*. arXiv:2509.15544v1. Theorems 1.5 and 1.7. https://arxiv.org/abs/2509.15544v1.

[8] Supplied project manuscript. *Higher-dimensional exponential metrics: A proposed axiomatic existence and uniqueness proof*. File `higher-dimensional-lqg-metric-revised.pdf`, in the supplied *LQG Results, 2026-10-17 to 19* archive. Theorems 1.2 and 1.4, Corollaries 10.2 and 10.4, and Lemma 10.3. The characterization is accepted as a standing project input in this paper.

[9] Supplied project manuscript. *First-order fluctuations of small-parameter exponential metrics*. File `unified-first-order-exponential-metrics.pdf`, in the same archive. Section 3, especially (3.20)–(3.21) and Proposition 3.2; also the display preceding (7.4). Its higher-dimensional assumption is the strong characterization in [8].

[10] Supplied project manuscript. *Almost sure convergence of higher-dimensional LFPP: An upgrade from convergence in probability*. File `higher_dimensional_lfpp_almost_sure.pdf`, in the same archive. Assumption 1.1 and Theorem 1.2. The parameter is fixed in that theorem.